

Forecasting Methods in Finance

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Abstract

Our review highlights some of the key challenges in financial forecasting problems and opportunities arising from the unique features of financial data. We analyze the difficulty of establishing predictability in an environment with a low signal-to-noise ratio, persistent predictors, and instability in predictive relations arising from competitive pressures and investors' learning. We discuss approaches for forecasting the mean, variance, and probability distribution of asset returns. Finally, we discuss how to evaluate financial forecasts while accounting for the possibility that numerous forecasting models may have been considered, leading to concerns of data mining.

1. INTRODUCTION

Finance is focused on intertemporal decision making under uncertainty, so forecasts of unknown future outcomes are integral to several areas of finance. Asset pricing requires forecasts of future cash flows, payoffs, and discount rates. Risk management relies on forecasts of variances and covariances of returns on portfolios that frequently comprise large numbers of assets. Countless studies in corporate finance analyze firms' capital budgeting decisions, which in turn depend on projected cash flows and firms' forecasts of the costs and benefits of issuing debt and equity. A large literature in banking analyzes the possibility of runs, which reflects not only investors' forecasts of a bank's solvency and liquidity but also their expectation of other agents' (depositors') decisions on whether to run or stay put.

While economic and financial forecasting share many methods and perspectives, some important features help differentiate the two areas. First, competitive pressures and market efficiency mean that the signal-to-noise ratio in many financial forecasting problems—particularly predictability of asset returns—is very low compared to standard forecasting problems in macroeconomics, in which the presence of a sizeable persistent component makes forecasting easier. The presence of weak predictors with low predictive power and the resulting importance of parameter estimation error are, therefore, the norm rather than the exception in financial forecasting.

Second, and related to the first point, fierce competition among asset managers in the financial markets means that predictable patterns in asset returns can be expected to self-destruct as a result of investors' attempts to exploit predictability and the resulting adjustment in prices. The possibility of readily trading on price forecasts makes the scope for feedback effects from forecasts to actual outcomes stronger in finance than in other areas of economics. Model instability is therefore particularly important to financial forecasting.

Third, overfitting and issues related to data mining have increasingly become a concern in financial forecasting due to the ease with which numerous forecasting models can be fitted to a given data set and the difficulty of generating new and genuinely independent data sets on which to test forecasting performance. In particular, how should the performance of a forecasting model be evaluated when this model is selected as the best performer among a larger set of competing specifications? This situation generates a multiple hypothesis testing problem that, if not accounted for, can lead to findings of spurious predictability patterns and serious distortions in inference.

Fourth, while volatility forecasting also features prominently in forecasting of macroeconomic variables—indeed, the original application of autoregressive conditional heteroskedasticity (ARCH) models was to UK inflation (Engle 1982)—it is more central to finance. This is particularly true in the area of risk management, which can entail forecasting the correlations between very large sets of variables, giving rise to high-dimensional forecasting problems. Moreover, access to high-frequency data, sampled every few seconds during trading sessions for the most liquid assets, means that measures of realized variances can be constructed and used to forecast future risks. This type of data does not, as yet, have obvious counterparts in economics where measurements tend to be conducted at a lower frequency.

Fifth, the presence of markets for derivatives, such as options or credit default swaps, means that risk-neutral densities can be constructed under no-arbitrage conditions and used to forecast the probability distribution of asset prices. Once converted into physical probability distributions, such density estimates can be combined with forecasts obtained from other sources. Using options data in this manner introduces a host of complexities, however, related to having limited cross-sectional data on liquid traded options.

Sixth, financial forecasting problems often involve well-defined loss functions, leading to optimization problems such as maximizing the expected utility from trading for an investor with mean-variance or power utility. In turn, this involves forecasting the probability distribution of

portfolio payoffs or particular moments of this distribution. Given such utility functions, it is now routine to evaluate forecasting performance using economic measures such as certainty equivalent returns or average realized utilities from investment strategies based on a sequence of forecasts.

A variety of methods have been—or have the potential to be—used to deal with these challenges in financial forecasting. For example, methods for dealing with weak predictors and parameter estimation error such as forecast combination and, more broadly, ensemble forecasting methods developed in machine learning are beginning to find more widespread use. Forecasting methods that take advantage of constraints from economic theory by, e.g., using filtering methods to back out persistent components in expected returns and expected dividend growth or imposing bounds on the conditional Sharpe ratio, have also shown promise. Our review discusses these and other strategies for improving financial forecasting performance.

Our review proceeds as follows. Section 2 introduces the basic return predictability problem. Section 3 discusses challenges encountered in financial forecasting problems, including weak predictors (low signal-to-noise ratios), persistent predictors, model instability, and data mining. Section 4 discusses strategies for dealing with these challenges. Section 5 addresses volatility and density forecasting methods, while Section 6 discusses methods for evaluating financial forecasts, emphasizing the use of economic performance measures, and Section 7 concludes.

2. BASICS OF RETURN PREDICTABILITY

Let r_{t+1} denote the excess return on a risky asset held from period t to period $t+1$, net of a risk-free rate. Ignoring frictions due to transaction costs and restrictions on trading, under conditions of no arbitrage the following moment condition holds:

$$E_t[m_{t+1}r_{t+1}] = 0, \quad 1.$$

where m_{t+1} is the positively valued stochastic discount factor (pricing kernel) (see, e.g., Cochrane 2005) and $E_t[\cdot] = E[\cdot | \Omega_t]$ denotes conditional expectations given information at time t , Ω_t .

Equation 1 shows that the product of the pricing kernel and excess returns is a martingale difference sequence and so has mean zero, conditional on the filtration generated by Ω_t . Solving for expected excess returns, we have

$$E_t[r_{t+1}] = \frac{-\text{cov}_t(r_{t+1}, m_{t+1})}{E_t[m_{t+1}]}, \quad 2.$$

where $\text{cov}_t(r_{t+1}, m_{t+1}) = E_t[(r_{t+1} - E_t[r_{t+1}])(m_{t+1} - E_t[m_{t+1}])]$ is the conditional covariance between r_{t+1} and m_{t+1} . This equation shows that predictability of excess returns is not ruled out by the absence of arbitrage. However, to be consistent with no-arbitrage conditions, any return predictability should reflect either time variation in the conditional covariance between excess returns and the stochastic discount factor, $\text{cov}_t(r_{t+1}, m_{t+1})$, or variation in the conditional expectation of the pricing kernel, $E_t[m_{t+1}]$.

A key challenge to interpretation of empirical evidence on return predictability is that the object that theory stipulates should be a martingale difference sequence, $m_{t+1}r_{t+1}$, is itself unobserved and model dependent. (For example, in a consumption-based asset pricing model, the pricing kernel reflects investors' intertemporal marginal rate of substitution between current and future consumption and, thus, depends on the assumed utility specification.) Hence, interpretations of return predictability should always bear in mind the joint hypothesis problem well-known from studies of market efficiency: Predictability tests are really joint tests of market efficiency and a correct specification of investor preferences. For example, stock returns may be predictably higher during recessions than during expansions simply because investors' marginal utility of consumption, and hence risk premia, may be higher during states with low growth.

By far the most commonly used prediction model in empirical studies is a simple linear specification for the equity premium:

$$r_{t+1} = \mu + \beta x_t + u_{t+1}, \quad 3.$$

where $x_t \in \Omega_t$ is a predictor variable known at time t . While the linear forecasting model in Equation 3 may appear to be at odds with the more general first-order form in Equation 1, in fact it can be derived under quite general conditions: Assuming an affine pricing kernel and cash flows that are formed as a linear combination of a finite-dimensional, stationary vector autoregression, Farmer, Schmidt & Timmermann (2017) show that Equation 3 can be derived from a log-linearized asset pricing model.

Further insights into the importance of forecasting for asset pricing can be gleaned from the log-linearized present value model of Campbell & Shiller (1988), which gives rise to the following approximate relation between the current log price, p_t ; forecasts of future log dividends, d_{t+1+j} ; and continuously compounded returns, r_{t+1+j} :

$$p_t = \frac{k}{1-\rho} + E_t \left[\sum_{j=0}^{\infty} \rho^j [(1-\rho)d_{t+1+j} - r_{t+1+j}] \right], \quad 4.$$

where k and ρ are constants arising from the log-linearization.

Computing the price of a perpetual asset such as a stock therefore requires forecasting an infinite stream of cash flows (log dividends, d_{t+1+j}) and discount rates (r_{t+1+j}). This complex task requires not only forecasting all future values of these variables themselves, but also forecasting the future values of any other variables used to predict cash flows and discount rates; this task is usually accomplished using vector autoregressions (VARs).

Letting Δd_{t+i} denote the log-dividend growth rate, it follows that surprises to returns are driven either by changes in expected future dividends or changes in expected future returns:

$$\begin{aligned} r_{t+1} - E_t[r_{t+1}] &= E_{t+1} \left[\sum_{j=0}^{\infty} \rho^j \Delta d_{t+1+j} \right] - E_t \left[\sum_{j=0}^{\infty} \rho^j \Delta d_{t+1+j} \right] \\ &\quad - \left(E_{t+1} \left[\sum_{j=1}^{\infty} \rho^j r_{t+1+j} \right] - E_t \left[\sum_{j=1}^{\infty} \rho^j r_{t+1+j} \right] \right). \end{aligned} \quad 5.$$

Noting that $E_{t+1}[\cdot]$ and $E_t[\cdot]$ represent forecasts computed conditional on information at time $t+1$ and time t , respectively, deviations in realized returns from their previously expected values must be driven by changes in dividend or return expectations. Importantly, all future values of these variables matter, so changes in forecasts of payoffs or discount rates at any future horizon should lead to corresponding changes in returns.

3. CHALLENGES TO FINANCIAL FORECASTING MODELS

Researchers attempting to identify predictability in asset returns face several challenges, which we describe in this section before discussing strategies for addressing such challenges in Section 4.

3.1. Challenge I: Weak Predictors

A very low signal-to-noise ratio in predictive return regressions is to be expected from market competition among asset managers and other investors who use vast resources to vie for higher returns.

One way to model the weak-predictor feature of financial return regressions is to assume that the coefficient of the time-varying predictor in Equation 3 is local to zero, i.e.,

$$\beta \propto \frac{b}{\sqrt{T}}, \quad 6.$$

for some constant, b , and a sample size, T . This approximation suggests that prediction models have limited power even in cases with a large T . Situations with weak predictors imply that parameter estimation error is roughly of the same order of magnitude as the signal embedded in the predictor, implying that conventional tests of predictive performance, such as that proposed by Diebold & Mariano (1995), will have little power to detect return predictability.

Bayesian methods have been used to counter the important effect of parameter estimation error on return forecasts. By shrinking the coefficient estimates toward a prior (usually centered on zero for the slope coefficients on time-varying predictor variables), these methods dampen the effect of estimation error on the forecasts. Even though such forecast methods may result in biased forecasts, they can exploit the bias–variance trade-off in such a way as to improve the forecasting models' mean squared error (MSE) performance.

From a variable selection perspective, weak predictors create a gray area where inclusion of individual predictors in return regressions is surrounded by considerable uncertainty that is unlikely to be conclusively resolved by conventional model selection methods.

The difficulty associated with establishing return predictability is countered by the fact that even small amounts of return predictability have the potential to translate into significant economic gains. Back-of-the-envelope calculations by Campbell & Thompson (2008) suggest that an (out-of-sample) R^2 value as low as 0.005 (one-half of 1%) in a monthly return regression could generate a 40% increase in the average portfolio excess return of a mean-variance investor with a modest degree of risk aversion. (This calculation ignores trading costs and slippage in prices as a result of market impact from trading.) For a discussion of this point, see also Zhou (2010).

3.2. Challenge II: Persistent Predictors

Many of the predictors used to forecast stock returns—notably valuation ratios such as the dividend yield, but also short-term interest rates or interest-rate spreads—are highly persistent. As pointed out by Stambaugh (1999), this can lead to biases in inference on the slope coefficient β in Equation 3 provided that the innovation in the predictor is strongly correlated with unexpected shocks to returns.

Suppose that the persistence in the predictor is captured by modeling this as a first-order autoregression:

$$x_t = \mu_x + \rho x_{t-1} + v_t, \quad |\rho| < 1, \quad v_t \sim (0, \sigma_v^2). \quad 7.$$

Assuming Gaussian innovations, Marriotti & Pope (1954) and Kendall (1954) show that there is a finite-sample bias in the estimated coefficient $\hat{\rho}$:

$$E[\hat{\rho} - \rho] \approx \frac{-(1 + 3\rho)}{T}.$$

Finally, suppose that $E(u_t | x_s, x_w) \neq 0$, $s < t \leq w$, so that the u_t residuals from the linear return regression in Equation 3 are correlated with past or future values of the predictor, x , i.e., $\sigma_{uv} = E[u_t v_t] \neq 0$. This condition obviously holds for predictors that have prices in the denominator, such as the dividend-to-price ratio.

Under these conditions, Stambaugh (1999, proposition 4 and corollary) shows that finite-sample biases in $\hat{\rho}$ translate into a finite-sample bias in $\hat{\beta}$:

$$E[\hat{\beta} - \beta] = \frac{\sigma_{uv}}{\sigma_v^2} E[\hat{\rho} - \rho] = \frac{-\sigma_{uv}}{\sigma_v^2} \frac{(1 + 3\rho)}{T} + O(T^{-2}). \quad 8.$$

Hence, if u_t and v_t are uncorrelated, there will not be a problem with a finite-sample bias in $\hat{\beta}$. Conversely, if $\sigma_{uv} \neq 0$, the bias can be substantial. For example, Stambaugh (1999) estimates a bias of about 0.40 in a regression of returns on the dividend yield from 1977–1996 ($T = 240$).¹

3.3. Challenge III: Model Instability

The same set of return predictor variables typically does not work for extended periods of time; the set of variables that get selected changes over time. No single prediction model is therefore clearly superior to other models. This complicates the selection of a forecasting model and produces an expected loss function that is a probability-weighted average across all models.

There is a particular reason why we would expect instability to affect financial forecasts. Asset returns depend on asset prices, which themselves reflect investors' expectations of future payoffs. A release of new public information that leads to a change in price forecasts can be expected to lead to simultaneous shifts in prices and, if shared by the broad market, will be incorporated into the market price with little delay.² If the information does not lead to any shift in risk premia, it cannot be expected to lead to improved forecasts of changes in prices or, more precisely, excess returns. Contrast this with new information about the state of the economy, e.g., the release of a jobs report, which can be expected to lead to more accurate forecasts of future economic growth, as it reveals more accurate information about the underlying state of the economy.

Consistent with the tendency for financial return models to self-destruct, the linear return prediction model in Equation 3 has been found to be unstable in many empirical studies. Paye & Timmermann (2006) and Rapach & Wohar (2006) test for parameter stability and find that the null of stability is rejected for the most commonly used predictor variables using returns data from the United States and a range of international stock markets.³ Pástor & Stambaugh (2001) use a Bayesian approach to find breaks in the relation between expected returns and return volatility (see also Maheu & McCurdy 2009).⁴

Farmer, Schmidt & Timmermann (2017) present evidence from nonparametric regressions that suggests that return predictability is concentrated in a small number of pockets with no significant predictability in the majority of the sample. As some of these pockets are short-lived, lasting a few weeks, this poses a particular challenge for real-time detection of predictability and attempts at exploiting such pockets to generate improved statistical and financial performance.

¹Stambaugh (1999) demonstrates that Bayesian inference can yield sharper results, although such inference will depend on the priors, assumptions about the initial observation (fixed or stochastic), and stationarity assumptions for the predictor.

²McLean & Pontiff (2016) find evidence that a range of stock market anomalies, i.e., signs of mispricing, tend to vanish after they are published in academic journals. They document a 32% average reduction in anomalies due to publication-informed trading. Moreover, the postpublication reduction in the magnitude of such anomalies appears to be greater in cases with larger in-sample returns and for stocks that have less idiosyncratic risk and are more liquid. These anomalies are easier to detect and to exploit through trading strategies, so we would expect them to disappear more rapidly once their existence becomes more broadly known.

³Some of this instability may be related to state-dependent forecasting performance. Rapach, Strauss & Zhou (2010) and Dangl & Halling (2012) find evidence that return predictability is strong during economic recessions but much weaker during expansions.

⁴For an extensive review of evidence on model instability, see Rossi (2013a). Rossi (2013b) reviews the track record of exchange-rate forecasting models and discusses how these are affected by model instability.

Conventional time-series regressions generally have a limited ability (weak power) to detect breaks in the regression parameters in real time, i.e., without lengthy delays, as only few postbreak observations are available. This problem is exacerbated in cases with weak predictors whose effect gets veiled by parameter estimation error: Detecting a shift in the parameter of a variable that only possesses weak predictive power—and whose inclusion in the forecasting model is questionable—is more difficult than if the variable had a large and highly significant effect. To put it bluntly, accurate detection of shifts to the parameter of a variable that generates an R^2 of less than 1% is always going to be difficult.

While it is a difficult task to detect breaks in financial forecasting models, converting such evidence into more accurate forecasts is even more challenging. This holds regardless of whether a break is detected through some econometric test or is based on subjective beliefs. For example, suppose it is believed that the election of Donald Trump as US president led to a change in forecasting models, e.g., for corporate cash flows as a result of changes in expectations of economic growth or prospects for tax reductions. In the immediate aftermath of Trump's election in November 2016, investors would have had very few data points from the new administration and so would not have been able to accurately estimate the parameters of the model. If not properly managed, this could result in erratic forecasts dominated by estimation error in the early stages after a break.

3.4. Challenge IV: Data Mining and Overfitting

Data mining is a concern that affects many forecasting problems. In its simplest form, it is closely linked to overfitting. To illustrate the problem, consider two linear forecasting models, the first of which (M_1) uses a set of predictors, $\mathbf{x}_{1t}$, while the second model (M_2) uses $\mathbf{x}_{1t}$ in addition to a set of predictors, $\mathbf{x}_{2t}$,

$$\begin{aligned} M_1 : y_{t+1} &= \beta_1' \mathbf{x}_{1t} + \varepsilon_{1t+1}, \\ M_2 : y_{t+1} &= \beta_{21}' \mathbf{x}_{1t} + \beta_{22}' \mathbf{x}_{2t} + \varepsilon_{2t+1}. \end{aligned} \quad 9.$$

Suppose that the coefficient estimates for these models are obtained by ordinary least squares so that

$$\begin{aligned} \hat{\beta}_1 &= \arg \min_{\beta_1} T^{-1} \sum_{t=0}^{T-1} (y_{t+1} - \beta_1' \mathbf{x}_{1t})^2, \\ \hat{\beta}_2 &= \arg \min_{\beta_2} T^{-1} \sum_{t=0}^{T-1} (y_{t+1} - \beta_2' \mathbf{x}_t)^2, \end{aligned} \quad 10.$$

where $\beta_2 = (\beta_{21}' \ \beta_{22}')'$ and $\mathbf{x}_t = (\mathbf{x}_{1t}' \ \mathbf{x}_{2t}')'$. This results in two forecasts, $y_{1t+1|t} = \hat{\beta}_1' \mathbf{x}_{1t}$ and $y_{2t+1|t} = \hat{\beta}_2' \mathbf{x}_t$.

Suppose that forecasting performance is measured using the same sample, $\{y_{t+1}\}_{t=0}^{T-1}$, that is used to estimate the parameters in Equation 10. Because M_2 nests M_1 , M_2 will always provide a (weakly) better in-sample fit:

$$T^{-1} \sum_{t=0}^{T-1} (y_{t+1} - \hat{\beta}_{21}' \mathbf{x}_{1t} - \hat{\beta}_{22}' \mathbf{x}_{2t})^2 \leq T^{-1} \sum_{t=0}^{T-1} (y_{t+1} - \hat{\beta}_1' \mathbf{x}_{1t})^2, \quad 11.$$

where equality holds only if the second set of predictors, $\mathbf{x}_{2t}$, are orthogonal to the forecast errors from the first model, $(y_{t+1} - \hat{\beta}_1' \mathbf{x}_{1t})$. This means that the bigger model (M_2) (almost) always comes out on top, with the lower in-sample MSE. This conclusion holds even if the bigger model can

be expected to perform worse in the population than the small model (M_1), perhaps due to its inclusion of redundant predictors. In other words, Equation 11 can hold simultaneously with

$$E \left[(y_{t+1} - \hat{\beta}_{21} \mathbf{x}_{1t} - \hat{\beta}'_{22} \mathbf{x}_{2t})^2 \right] > E \left[(y_{t+1} - \hat{\beta}'_1 \mathbf{x}_{1t})^2 \right], \quad 12.$$

where $E[\cdot]$ denotes population expectation. This overfitting effect makes data mining tempting, but also means that great care must be exercised when evaluating any improvements in forecasting performance resulting from the inclusion of additional predictor variables.

Ferson, Sarkissian & Simin (2003) find that the effect of data mining for predictor variables can be exacerbated in a setup with persistent predictors, which introduces a spurious regression bias. Spurious regression biases can arise if the return-generating process contains a highly persistent expected return component that is correlated in finite samples with similarly persistent predictor variables. Data mining gets reinforced by this effect, as the predictors that appear to be the best ones, and thus produce the highest R^2 values, may simply be maximizing the spurious regression bias.

Data mining concerns do not arise only in the context of time-series prediction models. Harvey, Liu & Zhu (2016) undertake a thorough study of the academic literature on factors that have been proposed to explain cross-sectional variation in expected returns. To deal with the multiple hypothesis testing issues that arise when a large number of factors need to be considered, they suggest that new factors need to generate t -statistics above 3.0 in order to be statistically significant after accounting for the multiple hypothesis testing problem.

4. STRATEGIES FOR ADDRESSING THE CHALLENGES

This section describes a variety of strategies that have been used to address the challenges outlined in Section 3. We start with forecast combinations; turn to filtering (unobserved components) methods and approaches for capturing model change; and then discuss Bayesian methods, machine learning techniques, and theory-induced constraints on forecasting models.

4.1. Forecast Combination

Forecast combination methods have been used extensively in economic forecasting but are less widespread in financial forecasting. This is somewhat surprising, given that a large literature has established the substantial benefits of combining forecasts in a wide set of areas (see, e.g., Clemen 1989, Timmermann 2006).

Why combine financial forecasts? One answer is that forecasters often employ many models with similar predictive performance, making it difficult to identify a single, superior model. Another reason lies in state-dependent forecasting performance: Certain models may work well under some market conditions but not in others, and it can be difficult to tell, *ex ante*, which conditions will prevail in the future. Alternatively, the forecasting environment may simply be unstable, rendering individual forecasting models' past track records unreliable for their future performance. A third reason is simply that of diversification: All models are misspecified, and combining forecasts, e.g., by using an equal-weighted average of forecasts, has the effect of diversifying across model uncertainty.⁵

⁵ As a case in point, the accuracy of individual forecasts of asset returns is often reduced by the effects of estimation error. One reason forecast combination succeeds in producing better return forecasts is that it diversifies the effect of estimation errors when these are not too strongly correlated across forecasting models.

To see how forecast combination works, consider a vector of n individual forecasts of some variable y , $\mathbf{f}_{t+1|t} = (f_{1t+1|t}, f_{2t+1|t}, \dots, f_{nt+1|t})'$, where $f_{jt+1|t}$ is the one-step-ahead forecast of y_{t+1} given information at time t , Ω_t , generated by the j -th model. The simple equal-weighted forecast combination takes the form

$$f_{t+1|t}^{\text{EW}} = \frac{1}{n} \sum_{j=1}^n f_{jt+1|t}. \quad 13.$$

This $1/n$ strategy has proven highly successful in empirical applications, including in forming portfolios (DeMiguel, Garlappi & Uppal 2007). There are no weights to estimate from the data in this combination scheme, and the weight on each forecast does not depend on the individual forecasts' past performance.

A more general approach to forecast combination that accounts for the individual models' past forecasting performance estimates the combination weights from a linear regression of the outcome, y_{t+1} , on the predictors:

$$y_{t+1} = \beta_0 + \sum_{j=1}^n \beta_j f_{jt+1|t} + \varepsilon_{t+1}. \quad 14.$$

An intercept term (β_0) is often included to ensure that the combined forecast is (unconditionally) unbiased. If each of the individual forecasts is believed to be unbiased, alternatively one can impose the constraints $\beta_0 = 0$ and $\sum_{j=1}^n \beta_j = 1$ in Equation 14 to preserve the unbiasedness of the combined forecast.

The work of Rapach, Strauss & Zhou (2010) is a notable exception to the relative shortage of papers in financial forecasting that use forecast combination methods. The authors fit univariate forecasting models to returns on the US stock market, using a set of predictors from Welch & Goyal (2008), and form an equal-weighted average of these forecasts. Suppose that the univariate return prediction models take the form

$$r_{t+1} = \gamma_{0j} + \gamma_{1j} x_{jt} + u_{jt+1}, \quad j = 1, \dots, n. \quad 15.$$

Estimates of each of these regressions can be used to generate a return forecast $\hat{r}_{jt+1|t} = \hat{\gamma}_{0j} + \hat{\gamma}_{1j} x_{jt}$, which in turn can be used to form an equal-weighted forecast

$$\bar{r}_{t+1|t}^{\text{EW}} = \frac{1}{n} \sum_{j=1}^n \hat{r}_{jt+1|t}. \quad 16.$$

Empirically, Rapach, Strauss & Zhou (2010) find that an equal-weighted combination of quarterly forecasts from 15 univariate models of the form in Equation 15 produces significantly more accurate forecasts than those from a model that assumes a constant equity risk premium and so imposes $\gamma_{1j} = 0$ in Equation 15. (It appears to be less important whether the mean, median, or a trimmed mean of the forecasts is used to compute the forecast combination.)

Elliott, Gargano & Timmermann (2013) generalize the approach of Rapach, Strauss & Zhou (2010) to complete subset regressions that use equal-weighted averages of all forecasting models that include a fixed number (k) of predictor variables. The univariate case with a single predictor proposed by Rapach, Strauss & Zhou is a special case ($k = 1$). [From a theoretical perspective, Elliott, Gargano & Timmermann (2015) show that the complete subset regression combination achieves variance reduction relative to a "kitchen sink" approach that includes the full list of predictor variables.] Elliott, Gargano & Timmermann (2013), using a data set similar to that of Rapach, Strauss & Zhou, empirically find that equal-weighted combinations of forecasts from return prediction models that include a small set of predictors—often two or three—perform

notably better than forecasts from individual forecasting models and also are better than univariate forecasts such as those considered by Rapach, Strauss & Zhou.

An alternative to averaging forecasts from a set of parsimonious models is to reduce the dimensionality of the set of predictors by assuming a latent factor structure that allows for the extraction of a small set of common factors (diffusion indexes) that drive variation in the predictor variables. Ludvigson & Ng (2007), Kelly & Pruitt (2013), and Neely et al. (2014), for example, use common components to predict returns.

4.2. Filtering and Unobserved Components Models

Investors' conditional expectations of future cash flows and discount rates are unobserved, and so cannot directly be used to compute prices in Equation 4 or to track return movements in Equation 5. Assuming that prices embody investors' forward-looking expectations, we can treat these expectations as latent variables, estimates of which can be computed using filtering methods under the assumption that the log-linearized present value model holds.

To see how this might work, let $\mu_t = E_t[r_{t+1}]$ be the expected return, while the expected dividend growth is denoted by $g_t = E_t[\Delta d_{t+1}]$. Following van Binsbergen & Koijen (2010), suppose that these follow AR(1) (first-order autoregressive) processes:

$$\mu_{t+1} = \delta_0 + \delta_1(\mu_t - \delta_0) + \varepsilon_{t+1}^\mu, \quad 17.$$

$$g_{t+1} = \gamma_0 + \gamma_1(g_t - \gamma_0) + \varepsilon_{t+1}^g. \quad 18.$$

In turn, the realized dividend growth is expressed as its expected value, g_t , plus an unexpected shock, ε_{t+1}^d :

$$\Delta d_{t+1} = g_t + \varepsilon_{t+1}^d. \quad 19.$$

Using the log-linearized present value model, van Binsbergen & Koijen (2010) show that the log price–dividend ratio takes the form

$$pd_t = A - B_1(\mu_t - \delta_0) + B_2(g_t - \gamma_0), \quad 20.$$

where A , B_1 , and B_2 are functions of the underlying parameters. Van Binsbergen & Koijen (2010) show that this setup yields a state-space system with two state equations (17 and 18) and two measurement equations (19 and 20), the latter without an error term.

Using this representation, estimates of expected returns and dividend growth rates can be computed using a Kalman filter. Van Binsbergen & Koijen (2010) find empirically that such filtered estimates have significant predictive power over future returns and dividend growth. Interestingly, both dividend growth and expected returns contain persistent components, with stronger persistence estimated for the latter.

Pástor & Stambaugh (2009) develop an approach that also treats expected returns as a latent process. Their predictive system approach is composed of the following state-space system for observable returns, r_{t+1} ; an observable predictor, x_{t+1} ; and an unobserved expected return, μ_{t+1} :

$$\begin{aligned} r_{t+1} &= \mu_t + u_{t+1}, \\ x_{t+1} &= (1 - \alpha)E_x + \alpha x_t + v_{t+1}, \\ \mu_{t+1} &= (1 - \beta)E_r + \beta \mu_t + w_{t+1}, \end{aligned} \quad 21.$$

where $(u_{t+1}, v_{t+1}, w_{t+1}) \sim N(0, \Sigma)$, with N being the standard normal distribution. For simplicity, we assume a univariate predictor, x_t , but Pástor & Stambaugh (2009) allow for multiple predictors. Assuming this joint dynamics, the linear regression in Equation 3 is misspecified if the predictor variable is imperfectly correlated with expected returns, i.e., unless $\text{corr}(v_{t+1}, w_{t+1}) = \pm 1$ and

$\alpha = \beta$. Conversely, with imperfect predictors, the full history of both the predictors and the returns will generally matter to the conditional expectation of returns and can be computed from a weighted sum of past innovations to these variables. Pástor & Stambaugh (2009) develop tests for serial correlation in predictive return regressions that can be used to detect the presence of imperfect predictors. Moreover, they show that the sign of the correlation between residuals of the expected and unexpected return equations can be used as a diagnostic for an imperfect predictor.

4.3. Tracking Model Change

Different strategies for dealing with model instability in the area of financial forecasting have been considered. A simple, yet general representation of model or parameter instability is to modify Equation 3 to allow for a time-varying slope coefficient:

$$r_{t+1} = \mu + \beta_t x_t + u_{t+1}. \quad 22.$$

Several approaches come under this umbrella. For example, we could specify the regression parameters as a (linear) function of observables, z_t , e.g., $\beta_t = \beta_0 + \beta_1 z_t$, in which case Equation 22 becomes

$$r_{t+1} = \mu + \beta_0 x_t + \beta_1 z_t x_t + u_{t+1} \quad 23.$$

(for applications to evaluation of fund performance, see, e.g., Ferson & Schadt 1996; Christopherson, Ferson & Glassman 1998).

Alternatively, one can assume that the parameters of the linear regression model in Equation 3 are latent variables that follow a process

$$\beta_t = \beta_{t-1} + \omega_t, \quad 24.$$

where $\omega_t \sim N(0, Q_t)$. This time-varying parameter model nests Equation 3 as a special case if $Q_t = 0$ for all t (for an application of this type of model to stock market return predictability, see Dangl & Halling 2012). Johannes, Korteweg & Polson (2014) propose a more complex version of the time-varying parameter model that allows for mean-reverting stochastic volatility and mean-reverting coefficients:

$$\begin{aligned} r_{t+1} &= \mu + \beta_0 x_t + \beta_{t+1} x_t + \sqrt{V_{t+1}^r} u_{t+1}, \\ \beta_{t+1} &= \rho \beta_t + \sigma_\beta \varepsilon_{\beta t+1}, \quad \varepsilon_{\beta t+1} \sim \text{iidN}(0, 1), \\ \log(V_{t+1}^r) &= \alpha_r + \beta_r \log(V_t^r) + \sigma_r \eta_{rt+1}, \end{aligned} \quad 25.$$

where iidN means identically and independently normally distributed.

Henkel, Martin & Nadari (2011) consider Markov switching vector autoregression (MSVAR) models to predict stock market returns. Let $\mathbf{y}_{t+1} = (r_{t+1} \mathbf{x}'_{t+1})'$, so that r_{t+1} is the first element of the $\mathbf{y}_{t+1}$ vector. Then the MSVAR(1) model can be written as

$$\mathbf{y}_{t+1} = \boldsymbol{\mu}_{s_{t+1}} + \mathbf{A}_{s_{t+1}} \mathbf{y}_t + \mathbf{u}_{t+1}, \quad 26.$$

where $\mathbf{u}_{t+1} \sim N(\mathbf{0}, \boldsymbol{\Sigma}_{s_{t+1}})$, $\boldsymbol{\mu}_{s_{t+1}}$ is now a column vector, $\mathbf{A}_{s_{t+1}}$ is a state-dependent VAR matrix, $s_{t+1} \in \{1, 2\}$, and $P(s_{t+1} = j | s_t = i) = p_{ij}$. Hence, there is a constant probability of staying in one of the two states, and both the mean and variance shift across the states. Note that the two states repeat in this model. This assumption is relaxed by Pettenuzzo & Timmermann (2011), who use the Chib (1998) change-point process to predict stock returns. In particular, they allow for K states, so that $s_t \in \{1, \dots, K\}$ in the sample $t = 1, \dots, T$, and they assume that the same

state never repeats, so that the state transition probability takes the form

$$\mathbf{P} = \begin{pmatrix} p_{11} & p_{12} & 0 & \cdots & 0 \\ 0 & p_{22} & p_{23} & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & & p_{KK} \end{pmatrix}. \quad 27.$$

This model characterizes shifts between the K states during the historical sample. To predict future out-of-sample returns, Pettenuzzo & Timmermann (2011) assume that the new parameters after a break are drawn from a meta-distribution, which means that they can forecast out-of-sample returns in a way that accounts for the possible effect of future breaks. This requires making assumptions about the underlying stochastic process that determines both the frequency and the magnitude of future breaks.

Empirically, many of these studies find that it is important to account for time variation in the parameters of return prediction models. Dangel & Halling (2012) find that the random walk coefficients model (Equation 24) quickly adapts to changes in the underlying return-generating process. Johannes, Korteweg & Polson (2014) also find that it is important to generalize the standard model to allow for time-varying coefficients and volatility clustering. Moreover, using the more general models to forecast returns appears to lead to improvements in portfolio performance. Henkel, Martin & Nadari (2011) find that return predictability in the stock market is closely linked to economic recession periods and that some predictors have predictive power only during recessions. Pettenuzzo & Timmermann (2011) find that accounting for the effects of past breaks and the possibility of future breaks in return prediction models can lead to significant economic gains when the resulting forecasts are used for asset allocation decisions.

A very different approach for handling model instability is to attempt to estimate the size and magnitude of recent breaks affecting the forecasting model and to use this information to compute improved forecasts. To dampen the effect of estimation error on forecasts computed on short data samples in the aftermath of a break, Pesaran & Timmermann (2007) propose using both pre- and postbreak data, perhaps downweighting prebreak data to reduce biases in the resulting parameter estimates.

Alternatively, Bayesian methods that pull parameter estimates toward an economically reasonable prior can be used. This approach is akin to shrinking the postbreak parameter estimates toward the prior and letting the degree of shrinkage taper off as more data points are cumulated in a new postbreak regime.

There are clear limitations to time-series approaches, which tend to have weak power to detect structural breaks. Moreover, some predictability may reflect low-frequency movements (e.g., business cycle variation) that affect the vast majority of firms, stocks, and countries and whose predictability can be difficult to ascertain given the paucity of observations at longer horizons.

Panel forecasting methods that exploit cross-sectional information to increase the power of the tests for instability can alternatively be used. Such methods can be expected to work better if the timing of breaks across cross-sectional units is not too heterogeneous, i.e., if breaks are common. Pooling cross-sectional and time-series information in this manner appears to be a promising avenue for handling model instability in financial forecasting models (for a recent application of this idea to a panel of stock market portfolios, see Smith & Timmermann 2017).

4.4. Bayesian Methods

Bayesian methods provide a coherent framework for handling parameter estimation error, model uncertainty, and model instability. This can perhaps best be illustrated in the context of the simple

linear regression model

$$r_{t+1} = \mu + \beta' \mathbf{x}_t + u_{t+1}, \quad u_{t+1} \sim N(0, \sigma_u^2), \quad t = 1, \dots, T-1, \quad 28.$$

where $\mathbf{x}_t$ is a vector of predictors. Denoting the conditioning data set by $\mathbf{Z}_t = \{r_\tau \mathbf{x}_\tau'\}$, $\tau = 1, \dots, t$, we can integrate out uncertainty about the underlying parameters μ , β , and σ_u^{-2} to obtain the posterior predictive density:

$$p(r_{t+1}|\mathbf{Z}_t) = \int p(r_{t+1}|\mu, \beta, \sigma_u^{-2}, \mathbf{Z}_t) p(\mu, \beta, \sigma_u^{-2}|\mathbf{Z}_t) d\mu d\beta d\sigma_u^{-2}. \quad 29.$$

Parameter uncertainty can be handled in a two-step procedure. In the first step, values of the parameters μ , β , and σ_u^{-2} are drawn from $p(\mu, \beta, \sigma_u^{-2}|\mathbf{Z}_t)$ using a Gibbs sampler. Under a set of standard normal-gamma priors, draws from the joint posterior distribution iterate forth and back between the distributions for the mean parameters (μ, β) and the precision parameter (σ_u^{-2}) . Given these parameter values, in the second step, draws from the outcome distribution are simple to implement because, conditional on the first-step parameter values,

$$r_{t+1}|\mu, \beta, \sigma_u^{-2}, \mathbf{Z}_t \sim N(\mu + \beta' \mathbf{x}_t, \sigma_u^{-2}). \quad 30.$$

Model uncertainty can also readily be handled in the Bayesian framework. Suppose there are n models of the form in Equation 28, i.e.,

$$r_{t+1} = \mu_j + \beta_j' \mathbf{x}_{jt} + u_{jt+1}, \quad u_{jt+1} \sim N(0, \sigma_{ju}^2), \quad t = 1, \dots, T-1, \quad 31.$$

where again $\mathbf{x}_{it}$ is a vector of predictors and $\mathbf{x}_{it} \neq \mathbf{x}_{jt}$ for $i \neq j$. Then we can generate a draw from the predictive density by weighting each of the n models by their posterior probabilities given the data, $p(M_j|\mathbf{Z}_t)$:

$$p(r_{t+1}|\mathbf{Z}_t) = \sum_{j=1}^n p(r_{t+1}|M_j, \mathbf{Z}_t) \times p(M_j|\mathbf{Z}_t). \quad 32.$$

Here, $p(r_{t+1}|M_j, \mathbf{Z}_t)$ is the posterior predictive density of model j , which can be computed using steps similar to those in Equations 29 and 30, whereas $p(M_j|\mathbf{Z}_t)$ can be computed using the steps described in Section 5.4.1.

As a simple application, suppose that we are considering models with different numbers of breaks or regimes. For example, $p(r_{t+1}|M_j, \mathbf{Z}_t)$ could correspond to a model with j breaks. Then we can use Equation 32 to integrate out uncertainty about the number of breaks, weighting the posterior predictive density of model M_j by its posterior probability.

4.5. Machine Learning Methods

New methods for conducting predictive analytics, often in the context of data sets that have large cross-sectional or time-series dimensions, or both, have been developed in recent years (for an excellent introduction, see Hastie, Tibshirani & Friedman 2009). These methods offer ways to flexibly estimate predictive models without imposing strong assumptions on their functional form while accounting for situations with large sets of predictors.

Key to the successful use of these methods appears to be control of their tendency to overfit due to their flexibility. The low signal-to-noise ratio encountered in many financial forecasting problems raises the risk that flexible fitting methods will simply fit noise in a given sample and be strongly affected by estimation error, more so than simpler linear models.

As an example of a recent machine learning method, consider regression trees that provide piecewise (constant) approximations to an unknown functional form by splitting the state space

into a set of J disjoint regions, $S_1, S_2, \dots, S_J$. Within each region, the function takes a constant value, μ_j , so that

$$\Upsilon(\mathbf{x}_t, \Theta_J) = \sum_{j=1}^J \mu_j I(\mathbf{x}_t \in S_j), \quad 33.$$

where $\mathbf{x}_t$ is a vector of predictor variables at time t and $\Theta_J = \{S_j, \mu_j\}$, for $j = 1, \dots, J$, is the set of parameters used to carve out the state space and the values of the constants, μ_j . $I(\mathbf{x}_t \in S_j)$ is an indicator variable that equals one if $\mathbf{x}_t \in S_j$ and otherwise equals zero. Finding the optimal way to discretize the sample space into the J regions is the tricky part, particularly if the dimension of the vector of predictor variables, $\mathbf{x}_t$, is high. In contrast, once the boundaries for the state space have been determined, the values of μ_j are easy to estimate under conventional loss functions such as MSE or mean absolute error (MAE).

Individual regression trees can be boosted by fitting successive trees to the residuals that remain after previous rounds of tree fitting. Specifically, boosted regression trees proceed iteratively by partitioning the data into subregions, given the preceding splits of the data. Trees are added with the objective of obtaining a better fit in regions that were poorly fitted by the initial trees. With B such boosting steps, a boosted regression tree can be obtained,

$$f_B(\mathbf{x}_t) = f_{B-1}(\mathbf{x}_t) + \Upsilon(\mathbf{x}_t, \Theta_{J,B}) = \sum_{b=1}^B \Upsilon(\mathbf{x}_t, \Theta_{J,b}), \quad 34.$$

where $\Upsilon(\mathbf{x}_t, \Theta_{J,b})$ is the regression tree fitted in the b -th iteration. Parameter estimates of these trees can be obtained as the solution to an optimization problem of the form

$$\hat{\Theta}_{J,b} = \arg \min_{\Theta_{J,b}} \left\{ y_{t+1} - [f_{b-1}(\mathbf{x}_t) + \Upsilon(\mathbf{x}_t, \Theta_{J,b})]^2 \right\}, \quad 35.$$

assuming squared error loss.

Ensemble learning methods can be employed to address some of these concerns. To reduce the risk of overfitting, shrinkage can be used so that each tree contributes only a small amount to the overall fit. For example, using a value of $\lambda = 0.001$, Equation 34 can be replaced by

$$f_B(\mathbf{x}_t) = f_{B-1}(\mathbf{x}_t) + \lambda \Upsilon(\mathbf{x}_t, \Theta_{J,B}). \quad 36.$$

Replacing the conventional squared error loss function in Equation 35 with a MAE loss function $T^{-1} \sum_{t=0}^{T-1} |y_{t+1} - f_B(\mathbf{x}_t)|$ can also help to provide more robust results, as the tendency to fit outliers is reduced by this loss function. Finally, subsampling methods have been shown to improve forecasting performance.

One advantage of methods such as boosted regression trees over conventional nonparametric methods is that they are less subject to the curse of dimensionality and, in fact, can be used to handle cases with large-dimensional predictors. Of course, this ability comes at a cost. First, there is no guarantee that the method will identify a global optimum in a high-dimensional parameter optimization problem. Second, this flexibility means that the risk of overfitting can be quite large. For a recent application of these methods to forecasting stock returns, see Rossi & Timmermann (2015).

Although machine learning methods will undoubtedly be used extensively to mine financial data sets in the hope of finding empirical regularities that can be used in trading algorithms, it is important to be aware of their limitations. Several nonparametric and semiparametric methods (e.g., sieve estimation) have been accessible to researchers for a long time and have not yet become part of applied financial researchers' toolkit. Often, financial data sets have a limited time-series span that can be dominated by rare events, such as the global financial crisis of 2007–2008, in which

relationships between predictors and the dependent variable may become unstable. While model instability will have adverse effects on any forecasting approach, flexible methods that require large samples to obtain accurate estimates of the conditional mean function can be expected to be particularly strongly affected by shifts in the underlying data-generating process.

4.6. Exploiting Information from Economic Theory to Restrict Forecasts

One way to address the adverse effects of estimation error and model uncertainty is to use economic theory or priors to restrict the functional form and/or the values of the parameters of the forecasting model. For example, Pástor & Stambaugh (2009) use informative priors to constrain the sign of the correlation between shocks to expected and unexpected returns in their predictive systems analysis. Similarly, no-arbitrage constraints are used to impose restrictions on dynamic Gaussian affine term structure models for bond returns by Ang & Piazzesi (2003) and Sarno, Schneider & Wagner (2016).

Pettenuzzo, Timmermann & Valkanov (2014) use economic constraints to modify the posterior distribution of the parameters of the linear return regression in Equation 3 in a way that allows the return prediction model to learn from the data. They consider two types of constraints, namely a constraint that imposes nonnegative equity premia and a constraint that bounds the conditional Sharpe ratio and incorporates time-varying volatility in the predictive regression.

Note that the linear regression model in Equation 3 does not rule out negative equity premium forecasts if unconstrained parameter estimates $\hat{\mu}, \hat{\beta}$ are used to generate forecasts; i.e., we could have, for a univariate predictor, x_τ ,

$$\hat{r}_{\tau+1|\tau} = \hat{\mu} + \hat{\beta}x_\tau < 0, \quad \tau = 1, \dots, t-1. \quad 37.$$

Such negative forecasts of excess returns may not seem reasonable, as we would expect risk-averse investors to command a positive risk premium for holding the stock market portfolio. To address this point, following Campbell & Thompson (2008), suppose that the equilibrium equity premium is expected to be nonnegative:

$$E[r_{\tau+1}|\Omega_\tau] = \mu + \beta x_\tau \geq 0, \quad \tau = 1, \dots, t. \quad 38.$$

Noting that the constraint in Equation 38 must hold at all points in time, this yields t constraints in a sample with t observations.

Alternatively, we can impose bounds on the conditional Sharpe ratio, i.e., the ratio of the conditional mean over the conditional volatility. To account for time-varying volatility, the predictive regression in Equation 3 can be modified as follows:

$$r_{\tau+1} = \mu + \beta x_\tau + \exp(b_{\tau+1})u_{\tau+1}, \quad u_{\tau+1} \sim N(0, 1), \quad 39.$$

where $b_{\tau+1}$ is the log-volatility of returns at $\tau + 1$. The dynamics of $b_{\tau+1}$ could, for example, be specified as a random walk model, $b_{\tau+1} = b_\tau + \xi_{t+1}$, with $\xi_{t+1} \sim N(0, \sigma_\xi^2)$.

Bounds on the conditional Sharpe ratio, $SR_{\tau+1|\tau} = (\mu + \beta x_\tau) / \exp(b_\tau + 0.5\sigma_\xi^2)$, take the form

$$SR^l \leq SR_{\tau+1|\tau} \leq SR^u, \quad \tau = 1, \dots, t. \quad 40.$$

The bounds in Equation 40 indirectly restrict the parameters $\theta = (\mu, \beta, \sigma_\xi^2)$ as well as the sequence of log-return volatilities $h^t \equiv \{b_1, b_2, \dots, b_t\}$. In practice, Pettenuzzo, Timmerman & Valkanov (2014) impose the bounds $SR^l = 0$ and $SR^u = 1$ on the annualized Sharpe ratio.

Pettenuzzo, Timmermann & Valkanov (2014) assume standard Gaussian priors on the regression parameters and use a Gibbs sampler to estimate the forecasting model in Equation 39. They find empirically that imposing economic constraints succeeds in reducing uncertainty about

model parameters and reduces the risk of selecting a poor forecasting model. When evaluated in an out-of-sample forecasting analysis, the constrained models improve on both statistical and economic measures of forecasting performance.

An alternative approach, advocated by Ferreira & Santa-Clara (2011), is to disaggregate returns into cash flow–related and price–related components that can then be predicted separately. Ferreira & Santa-Clara decompose the continuously compounded returns, r_{t+1} , into growth in the log price–earnings multiple, gm_{t+1} , growth in log earnings, ge_{t+1} , and the log-dividend yield, dp_{t+1} :

$$r_{t+1} = gm_{t+1} + ge_{t+1} + dp_{t+1}. \quad 41.$$

Using this sum-of-parts approach, Ferreira & Santa-Clara forecast the three return components assuming (a) no change in the price–earnings multiple, so $E[gm_{t+1}] = 0$ (the simplest model); (b) a 20-year moving average for growth in earnings; and (c) a random walk for the dividend yield. (Ferreira & Santa-Clara also consider alternative forecasting models, such as a linear regression that incorporates mean reversion in the price–earnings multiple.) Their empirical results suggest that forecasting the disaggregated return components in this manner works well out-of-sample. One reason this approach is found to work well is that it is akin to shrinkage toward current values (for the price–earnings multiple and the dividend yield) and a slowly moving estimate of earnings growth. These choices reduce the effect of estimation error on the forecasts, which is an important consideration.

4.7. Cross-Validation (Out-of-Sample) Methods

Overfitting and data mining issues arise because of the correlation between the forecast error and the estimation error, which causes the estimated MSE in Equation 11 to be an underestimate of the (true) performance we would expect in a new sample. Cross-validation methods can be used to remove this correlation. Although these methods use the full data set, they avoid using the same data for model fitting (and selection) and for forecast evaluation.

One version of cross-validation that is particularly popular in finance (and in economics more broadly) is to split the sample $t = 1, \dots, T$ into an in-sample portion, $t = 1, \dots, R$ ($R < T$), used for model estimation and selection and an out-of-sample portion, $t = R + 1, \dots, T$, used for forecast evaluation.

To see how out-of-sample forecasting methods can help reduce concerns of overfitting, consider the simple linear regression model

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \mathbf{I}_T), \quad 42.$$

where $\mathbf{Y}$ and $\boldsymbol{\varepsilon}$ are $T \times 1$ vectors of the dependent variable and regression residuals, respectively, and the predictors have been stacked into a $T \times k$ matrix ($\mathbf{X}$) that has been rotated and standardized so that $\mathbf{X}'\mathbf{X} = \mathbf{I}_k$. In this setting, Hansen & Timmermann (2015a) show that the in-sample (IS) residual sum of squares (RSS) from the forecasting model is given by

$$RSS_{IS} = \boldsymbol{\varepsilon}'\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}'\mathbf{X}\mathbf{X}'\boldsymbol{\varepsilon}. \quad 43.$$

To capture the separation of the estimation and evaluation samples used by out-of-sample methods, suppose that $\boldsymbol{\beta}$ is estimated from an independent sample $\tilde{\mathbf{Y}} = \mathbf{X}\boldsymbol{\beta} + \tilde{\boldsymbol{\varepsilon}}$, with $\tilde{\boldsymbol{\varepsilon}} \sim N(\mathbf{0}, \mathbf{I}_T)$ being independent of $\boldsymbol{\varepsilon}$. The resulting out-of-sample (OoS) residual sum of squares is then given by

$$RSS_{OoS} = \boldsymbol{\varepsilon}'\boldsymbol{\varepsilon} + \tilde{\boldsymbol{\varepsilon}}'\mathbf{X}\mathbf{X}'\tilde{\boldsymbol{\varepsilon}} - 2\boldsymbol{\varepsilon}'\mathbf{X}\mathbf{X}'\tilde{\boldsymbol{\varepsilon}}. \quad 44.$$

When evaluating the in-sample and out-of-sample measures of forecasting performance in Equations 43 and 44, note that the RSS computed using the true value of $\boldsymbol{\beta}$ is $\boldsymbol{\varepsilon}'\boldsymbol{\varepsilon}$. Labeling this

value as RSS_{true} , using Equation 43, the overfit from using the in-sample measure of forecasting performance is given as follows (Hansen & Timmermann 2015a):

$$RSS_{\text{true}} - RSS_{\text{IS}} = \varepsilon' \mathbf{X} \mathbf{X}' \varepsilon \sim \chi_k^2. \quad 45.$$

This result makes intuitive sense: The more predictors are included in the forecasting model, the greater the potential for overfit, and the degree of overfit grows in direct proportion with the number of predictor variables, k . The key is to notice that the quadratic form $\varepsilon' \mathbf{X} \mathbf{X}' \varepsilon$ gets subtracted from the RSS, so that estimation error wrongly contributes to overfitting.

In contrast, using Equation 44, the overfit associated with the out-of-sample forecast evaluation scheme is (Hansen & Timmermann 2015a)

$$RSS_{\text{true}} - RSS_{\text{OoS}} = -\tilde{\varepsilon}' \mathbf{X} \mathbf{X}' \tilde{\varepsilon} + 2\varepsilon' \mathbf{X} \mathbf{X}' \tilde{\varepsilon}. \quad 46.$$

Recalling that ε and $\tilde{\varepsilon}$ are independent, so the second term has zero mean, we see that the expectation of the overfit is negative and equals $-\chi_k^2$. Hence, parameter estimation error serves to reduce the out-of-sample forecasting performance, as we would expect.

Besides reducing the problem arising from individual models' tendency to overfit, out-of-sample forecast evaluation tests have the advantage that they can help uncover periods in which a forecasting model produces accurate forecasts as well as periods in which it fails to do so (Giacomini & Rossi 2009, 2010; Rossi 2013a,b). This is particularly useful in financial forecasting, where we would expect forecasting models to work well only during limited periods of time due to competitive market pressure. Closely related to this, if the out-of-sample forecasts are careful in using only information and forecasting methods that are available in real time, evidence from such tests can be used to test market efficiency.

While out-of-sample forecast evaluation methods can be an effective way to reduce overfitting, they are no panacea and also introduce new problems. First, nothing prevents individual researchers—or a group of researchers, each of whom only considers a single model—from experimenting with a multitude of forecasting models. Picking the model with the best out-of-sample forecasting performance distorts inference if not properly accounted for. Data mining is not impossible in out-of-sample forecast experiments—it is merely more difficult than when conducted in-sample. As an illustration, simple calculations by Hansen & Timmermann (2015a) show that, for a regression model with $k = 4$ predictors, to achieve a rejection rate of 5% would take close to 5,000 different models in a sample with $T = 50$ observations. In stark contrast, the same rejection rate can be accomplished with a single model when based on in-sample forecasting performance.

Second, out-of-sample tests of forecasting performance can have substantially weaker power than in-sample tests. This is a straightforward implication of using fewer observations ($T - R$) to evaluate predictive accuracy than the corresponding in-sample tests (T). Inoue & Killian (2005) and Hansen & Timmermann (2015b) analyze the importance of this point. For nested regression models, Hansen & Timmermann (2015a) show that a standard test statistic for out-of-sample forecasting performance can be written as the difference between two Wald statistics of the same null hypothesis, namely a test that uses the full sample and a test that uses only a subsample. Subtracting a test statistic in this manner clearly reduces the power of the test and so results in less ability to identify predictability in out-of-sample experiments.

Third, the value of R , i.e., the choice of the sample split point that separates the in-sample and out-of-sample periods, may affect the results. Hansen & Timmermann (2012) analyze the effect of data mining over the sample split point. Let $\rho \in (0, 1)$ be the in-sample fraction of the data, so that $R_\rho = \lfloor \rho T \rfloor$ observations are used for parameter estimation, while $T - R_\rho$ observations are used for forecast evaluation. Hansen & Timmermann (2012) show that searching over the sample split point, ρ , can lead to severe distortions in inference if not accounted for

when evaluating predictive performance. In particular, they find that large values of ρ are more likely to be selected in experiments that choose ρ so as to maximize conventional test statistics of out-of-sample forecasting performance. This finding reflects the presence of the second term in Equation 46, which has zero mean but which has larger variance when the out-of-sample period is short and so is likely to result in higher rejection rates for tests of out-of-sample forecasting performance. This point can matter a great deal in practice: Hansen & Timmermann (2012) find that the rejection rate for an out-of-sample test with a nominal size of 5% can be quadrupled with only 3–4 predictor variables as a result of inflation in the test statistic induced by mining over the sample split point.

4.8. Accounting for the Multiple Hypothesis Testing Problem

Out-of-sample forecast evaluation methods can significantly reduce the tendency of models to overfit, which is associated with in-sample forecast evaluation. However, if many different models are being considered and this is not accounted for when evaluating the performance of the best forecasting model, severe distortions of inference on predictability may ensue.

This problem is analyzed by White (2000), who also provides an elegant solution to it. Suppose that there are k forecasting models, whose out-of-sample MSE performance is measured relative to some benchmark (model 0) so that the MSE differential of the benchmark relative to the k -th forecasting model is given by $\bar{d}_k = (T - R)^{-1} [\sum_{t=R}^{T-1} (y_{t+1} - f_{0t+1|t})^2 - (y_{t+1} - f_{kt+1|t})^2]$. Stacking the MSE differentials for the k models gives a vector of sample averages $\bar{\mathbf{d}} = (\bar{d}_1, \dots, \bar{d}_k)'$. Since $\bar{d}_k > 0$ indicates that the k -th forecasting model has outperformed the benchmark, the null hypothesis that none of the models is capable of beating the benchmark takes the form (White 2000)

$$H_0 : \max_{j=1, \dots, k} E[d_{jt+1|t}(\beta^*)] \leq 0. \quad 47.$$

Note that the null hypothesis is being evaluated at the probability limit of the parameter values β^* . If all forecasting models produce forecasts as accurate as those from the benchmark, the $\bar{\mathbf{d}}$ vector, as well as the maximum of this vector, has a mean of zero.

To evaluate if the null in Equation 47 is satisfied requires calculating the sampling distribution of the maximum of a k -dimensional vector of possibly correlated terms. To this end, White (2000) establishes conditions under which $\sqrt{(T - R)}\{\bar{\mathbf{d}} - E[d(\beta^*)]\}$ converges to a multivariate normal distribution with an unknown covariance matrix, $\mathbf{\Lambda}$. This insight reduces tests of the null in Equation 47 to the task of evaluating whether the performance of the best forecasting model, chosen from a set of k models, is better than one would expect by chance from picking the maximum value drawn from a k -dimensional normal distribution with covariance matrix $\mathbf{\Lambda}$.

To perform this skill-versus-luck calculation, White (2000) shows that a bootstrap that repeatedly draws samples from the underlying forecast data can be used to evaluate whether the performance of the best model in the actual data is genuinely better than what one would expect as a result of having picked such a model from a possibly large-dimensional vector of forecasts. Importantly, this approach sidesteps the need for estimating the covariance matrix, $\mathbf{\Lambda}$, which could be complicated in situations where k is large.

In an empirical application of the reality check bootstrap, Sullivan, Timmermann & White (1999) consider the performance of 7,846 technical trading rules applied to daily returns on stock market indexes or futures prices. Their list of technical trading rules includes filter rules, moving averages, support- and resistance-type rules, and channel breakouts. While the best technical trading rule generates performance that is highly statistically significant when evaluated in isolation, i.e., as if only a single model had been considered, Sullivan, Timmermann & White (1999) also

find that the best trading rule fails to outperform a buy-and-hold benchmark once one accounts for the fact that this model was selected as the best performer from a larger set. For an application of White's (2000) approach to modeling seasonality in the Treasury market, see Kamstra, Kramer & Levi (2015).

White's (2000) approach has been expanded in several interesting directions. For example, Romano & Wolf (2005) develop a stepwise approach that performs White's bootstrap iteratively so as to identify all models with superior performance over a benchmark, as opposed to only testing if there exists at least one model that beats the benchmark. Romano & Wolf (2005) provide conditions under which their stepwise approach can asymptotically select all superior models (and eliminate inferior models) subject to controlling the family-wise error rate, i.e., the probability of incorrectly identifying at least one forecasting model as being superior to the benchmark.

Hansen, Lunde & Nason (2011) develop a model confidence set approach that identifies the subset of forecasting models that, at a given level of confidence, includes the best-performing forecasting models, i.e., models whose performance is not dominated either by any other models in a pairwise test or when measured relative to the average performance of other models. Their approach involves an equivalence test for pairwise testing that model performance is identical, as well as an elimination rule for determining which model, if any, gets discarded from the model confidence set in a given step.

Both of these approaches hold considerable promise for applications in financial forecasting problems in which it is important to deal with the multiple hypothesis testing problem.

5. VOLATILITY AND DENSITY FORECASTING

A vast empirical literature shows evidence of persistence in the volatility of returns on a variety of asset classes (e.g., stocks, bonds, commodities, and currencies) and at different frequencies (e.g., daily, weekly, and monthly). Such persistence implies that time variation in return volatility is predictable. (For an excellent review of volatility forecasting methods, see Andersen et al. 2006.)

Early empirical evidence emanated from the ARCH and generalized autoregressive conditional heteroskedasticity (GARCH) models proposed by Engle (1982) and Bollerslev (1986), respectively. GARCH models take the form

$$\begin{aligned}\varepsilon_{t+1} &= \sigma_{t+1|t} \eta_{t+1}, \quad \eta_{t+1} \sim \text{iidN}(0, 1), \\ \sigma_{t+1|t}^2 &= \omega + \sum_{i=1}^p \beta_i \sigma_{t+1-i|t-i}^2 + \sum_{i=1}^q \alpha_i \varepsilon_{t+1-i}^2.\end{aligned}\quad 48.$$

The recursive form of this specification means that it is straightforward to generate volatility forecasts, iterating to obtain a forecast of the b -steps-ahead volatility $\sigma_{t+b|t}^2$ from the $(b-1)$ -steps-ahead forecast, $\sigma_{t+k-1|t}^2$.

The GARCH(1,1) model

$$\sigma_{t+1|t}^2 = \omega + (\alpha_1 + \beta_1) \sigma_{t|t-1}^2 + \alpha_1 \sigma_{t|t-1}^2 (\eta_t^2 - 1) \quad 49.$$

has proven highly successful in many empirical applications (Hansen & Lunde 2005). For this model, the persistence of the conditional variance can be measured by $\alpha_1 + \beta_1$, as $E_{t-1}[\eta_t^2 - 1] = 0$. It is not uncommon to find estimates for which $\hat{\alpha}_1 + \hat{\beta}_1$ is slightly below one, indicating a high degree of persistency, particularly at high frequencies. This is consistent with the presence of a strong predictive component in the second moment of the return distribution. Assuming that

$\alpha_1 + \beta_1 < 1$, the steady-state (average) variance from the GARCH process in Equation 49 is

$$E[\sigma_{t+1|t}^2] \equiv \sigma^2 = \frac{\omega}{1 - \alpha_1 - \beta_1},$$

and a forecast of the conditional one-period variance b periods from now, given current information, can be computed as

$$\sigma_{t+b|t}^2 \equiv E_t[\sigma_{t+b}^2] = \sigma^2 + (\alpha_1 + \beta_1)^{b-1}(\sigma_{t+1|t}^2 - \sigma^2), \quad b \geq 1.$$

Regime-switching models offer an alternative way to capture persistence in volatility. Consider this simple regime-switching model with a state-dependent mean and volatility:

$$r_{t+1} = \mu_{s_{t+1}} + \sigma_{s_{t+1}} \eta_{t+1}, \quad \eta_{t+1} \sim \text{iidN}(0, 1), \quad 50.$$

where $\mu_{s_{t+1}}$ and $\sigma_{s_{t+1}}$ are the mean and volatility in state s_{t+1} . As in the literature on return predictability, it is typically assumed that there is a small set of possible states, i.e., $s_{t+1} \in \{1, 2, \dots, K\}$ for a small value of K . For $K = 2$, there are only two possible states, often identified as high- and low-volatility states when these models are fitted to asset returns. Persistence in the level of volatility can be modeled by assuming that s_{t+1} follows a first-order homogeneous Markov chain

$$\Pr(s_{t+1} = j | s_t = i) = p_{ij}. \quad 51.$$

The Markov chain is persistent provided that $p_{11} + p_{22} > 1$ in the case with two states. Time variation in the persistence of return volatility can be captured by using dynamic state transition probabilities of the form

$$\Pr(s_{t+1} = j | s_t = i, z_t) = p_{ij}(z_t), \quad 52.$$

where $p_{ij}(z_t)$ can be specified as a logit or probit function to ensure that $p_{ij}(z_t) \in [0, 1]$. For further details on the properties and estimation of regime switching models, see Hamilton (1989).

GARCH and Markov switching models can be used to capture features of financial data such as fat tails (kurtosis) and skew. For example, time-varying skews can be incorporated into the GARCH model by allowing positive and negative shocks to have a different effect on future volatility, as captured by the threshold GARCH model of Glosten, Jagannathan & Runkle (1993):

$$\sigma_{t+1|t}^2 = \omega + \alpha_1 \varepsilon_t^2 + \lambda \varepsilon_t^2 I(\varepsilon_t < 0) + \beta_1 \sigma_{t|t-1}^2, \quad 53.$$

where $I(\varepsilon_t < 0)$ is an indicator variable that takes a value of one if $\varepsilon_t < 0$ and otherwise equals zero.

Similarly, the return distribution generated by the MS model in Equation 50 with two states will be skewed provided that $\mu_1 \neq \mu_2$.

GARCH and Markov switching specifications such as those in Equations 48 and 50 are adaptive in that they do not predict an initial large shock, i.e., an outlier in ε_{t+1} that is not preceded by a large value of ε_t . However, once such a shock has occurred, these models will elevate their forecast of the next periods' variance using a decay rate that reflects mean reversion, provided that $\alpha_1 + \beta_1 < 1$ in the GARCH model (Equation 49) or $p_{ii} < 1$ in the MS model (Equation 51). This property, coupled with the persistence of the underlying volatility process, accounts for the empirical success of these classes of models when used to model volatility of asset returns.

5.1. Realized Volatility

Realized volatility models make use of data sampled at a higher frequency than that typically used to estimate GARCH models. For example, a realized volatility model could be based on data

sampled every 5 minutes during the market opening hours of the trading day, whereas a GARCH model typically uses daily returns data. (For an analysis of how often data should be sampled when they are affected by market microstructure noise, see Ait-Sahalia, Mykland & Zhang 2005). To see how realized volatility models work, suppose that changes in log prices, dp_t , evolve according to an arithmetic random walk process with constant drift μ and time-varying volatility σ_t :

$$dp_t = \mu dt + \sigma_t dW_t, \quad 54.$$

where dW_t are increments to an underlying Wiener process. The volatility of asset returns over some interval $[t-1, t]$ is unobserved but can be approximated by sampling returns on a discrete grid of points $t-1 < \tau_0 < \tau_1 < \dots < \tau_N = t$. This gives rise to the following realized variance:

$$RV_t = \sum_{j=1}^N (dp_{\tau_j} - dp_{\tau_{j-1}})^2. \quad 55.$$

For a discussion of the properties of the estimated realized variance, such as consistency, see Hansen & Lunde (2011).

We would expect RV_t to be a somewhat noisy estimate of the underlying volatility for small values of N . However, because the realized variance can be viewed as a noisy proxy for the unobserved (underlying) volatility process, one can imagine using it as the basis for a forecasting equation,

$$RV_{t+1} = \alpha + \sum_{j=1}^J \beta_j RV_{t+1-j} + \varepsilon_{t+1}. \quad 56.$$

A notable example of this model that has been extensively used in empirical work is the HAR-RV model proposed by Corsi (2009), which involves projecting the daily integrated volatility ($\tilde{\sigma}_{t+1}^2$) on lagged values of daily (RV_t^d), weekly (RV_t^w), and monthly (RV_t^m) realized volatility estimates:

$$\tilde{\sigma}_{t+1}^2 = \alpha + \beta_d RV_t^d + \beta_w RV_t^w + \beta_m RV_t^m + \varepsilon_{t+1}, \quad 57.$$

where, for example, $RV_t^w = \frac{1}{5}(RV_t^d + RV_{t-1}^d + RV_{t-2}^d + RV_{t-3}^d + RV_{t-4}^d)$ is the weekly average realized volatility. Using past realized volatilities at different frequencies often improves the predictive power over a model that only includes a few lags of the daily realized volatility measure.

Alternatively, RV_t can be added as an additional covariate to GARCH models or modeled jointly with the log variance from a GARCH process to get a model such as

$$\begin{aligned} \log \sigma_{t+1|t}^2 &= \omega + \beta \log \sigma_{t|t-1}^2 + \gamma RV_t, \\ \log RV_t &= \delta_0 + \delta_1 \log \sigma_{t|t-1}^2 + \delta_2 \log RV_{t-1} + v_t, \end{aligned} \quad 58.$$

in the spirit of Hansen, Huang & Shek (2012). [Paye (2012) finds that univariate specifications of monthly stock market volatility such as Equation 56 are difficult to beat in out-of-sample forecast comparisons when compared to models that add a variety of macroeconomic predictor variables to the forecasting equation.]

High-frequency data on price movements give researchers the ability to better decompose total volatility into separate continuous volatility and jump volatility components. This can be important for forecasting purposes because the two components possess very different time-series properties. Notably, jump volatility tends to be much less persistent than continuous volatility. For examples of this approach, see Andersen, Bollerslev & Diebold (2007) and Patton & Sheppard (2015).

Some studies suggest an important role in volatility forecasting for implied volatility measures extracted from options data. For example, studies such as that of Blair, Poon & Taylor (2001) find

that, once information in implied volatilities is used to predict future volatility, there is little or no additional information in high-frequency return movements.

5.2. Stochastic Volatility

GARCH models such as that in Equation 48 assume that volatility is driven by past and current shocks to returns that are observable and, thus, can be estimated by maximum likelihood methods subject to assumptions on the value of the initial volatility state. Stochastic volatility models instead assume that the volatility process is affected by a sequence of volatility-specific shocks, ζ_t .

The basic stochastic volatility model takes the form

$$r_{t+1} = \mu + \exp(b_{t+1})\eta_{t+1}, \quad \eta_{t+1} \sim \text{iidN}(0, 1), \quad 59.$$

where the log volatility at time $t + 1$, b_{t+1} , can be modeled as a stationary, mean-reverting process:

$$b_{t+1} = \lambda_0 + \lambda_1 b_t + \zeta_{t+1}, \quad \zeta_{t+1} \sim \text{iidN}(0, \sigma_\zeta^2), \quad 60.$$

for $|\lambda_1| < 1$. The shocks η_τ and ζ_s are mutually independent for all values of τ and s . It is also not uncommon to impose that $\lambda_0 = 0$ and $\lambda_1 = 1$, which turns Equation 60 into a driftless random walk model.

Kim, Shephard & Chib (1998) develop methods for estimation of stochastic volatility models. The presence of a volatility-specific shock sequence ζ_{t+1} in Equation 60 means that conventional maximum likelihood methods cannot be used for estimation. Instead, the Gibbs sampler can be used to obtain draws from the joint posterior distribution $p(\mu, \mathbf{h}^t, \lambda_0, \lambda_1, \sigma_\zeta^{-2})$, where $\mathbf{h}^t = (b_1, \dots, b_t)$. With these in place, one can obtain draws from the predictive density

$$\begin{aligned} p(r_{t+1} | \mathbf{\Omega}_t) &= \int p(r_{t+1} | b_{t+1}, \mu, \mathbf{h}^t, \lambda_0, \lambda_1, \sigma_\zeta^{-2}, \mathbf{\Omega}_t) \\ &\quad \times p(b_{t+1} | \mu, \mathbf{h}^t, \lambda_0, \lambda_1, \sigma_\zeta^{-2}, \mathbf{\Omega}_t) \\ &\quad \times p(\mu, \mathbf{h}^t, \lambda_0, \lambda_1, \sigma_\zeta^{-2} | \mathbf{\Omega}_t) d\mu d\mathbf{h}^{t+1} d\lambda_0 d\lambda_1 d\sigma_\zeta^{-2}, \end{aligned} \quad 61.$$

where $p(r_{t+1} | b_{t+1}, \mu, \mathbf{h}^t, \lambda_0, \lambda_1, \sigma_\zeta^{-2}, \mathbf{\Omega}_t)$ is the predictive density conditional on the model parameters, including the value of the log volatility; $p(b_{t+1} | \mu, \mathbf{h}^t, \lambda_0, \lambda_1, \sigma_\zeta^{-2}, \mathbf{\Omega}_t)$ captures shifts in the future log volatility, b_{t+1} , away from the current log volatility, b_t ; and $p(\mu, \mathbf{h}^t, \lambda_0, \lambda_1, \sigma_\zeta^{-2} | \mathbf{\Omega}_t)$ measures the effect of parameter uncertainty. Equation 61 can be used to generate draws from the predictive density of returns that can, in turn, be used to compute optimal portfolio weights or to evaluate functions of the conditional return distribution.

As in the case of GARCH models or similar volatility specifications, observable predictors can also be added to the stochastic volatility model. For a mixed data sampling (MIDAS) approach that adds predictor variables to the stochastic volatility specification in Equation 60, see Pettenuzzo, Timmermann & Valkanov (2016); for broader summaries of MIDAS models, see Ghysels, Sinko & Valkanov (2007) and Andreou, Ghysels & Kourtellis (2011).

5.3. Multivariate Volatility Models and Copulas

Portfolio allocation and risk management problems involve modeling and forecasting the joint distribution of a possibly very large set of asset returns. Several approaches have been developed to capture predictable time variation both in the probability distribution of individual assets' returns and in their joint distribution. Consider how to model the returns on an $n \times 1$ vector of asset

returns, $\mathbf{r}_{t+1}$, using a conditional scale-location model:

$$\mathbf{r}_{t+1} \sim N(\boldsymbol{\mu}_t, \mathbf{H}_{t+1|t}). \quad 62.$$

The multivariate dynamic conditional correlation model of Engle & Sheppard (2001) and Engle (2002) takes the form

$$\mathbf{H}_{t+1|t} = \mathbf{D}_{t+1|t} \mathbf{R}_{t+1|t} \mathbf{D}_{t+1|t}, \quad 63.$$

where $\mathbf{D}_{t+1|t}$ is a diagonal $n \times n$ matrix with time-varying volatilities whose i -th diagonal element, $b_{iit+1|t}^{1/2}$, can be generated from a univariate GARCH process of the form

$$b_{iit+1|t} = \omega_i + \sum_{p=1}^{p_i} \alpha_{ip} \varepsilon_{iit+1-p}^2 + \sum_{q=1}^{q_i} \beta_{iq} b_{iit+1-q|t-q}. \quad 64.$$

The matrix $\mathbf{R}_{t+1|t}$ captures time-varying correlations and is modeled by Engle & Sheppard (2001) as

$$\mathbf{R}_{t+1|t} = (\mathbf{Q}_{t+1|t}^*)^{-1} \mathbf{Q}_{t+1|t} (\mathbf{Q}_{t+1|t}^*)^{-1}, \quad 65.$$

where $\mathbf{Q}_{t+1|t}^*$ is a diagonal matrix with individual elements $q_{iit+1|t}^{1/2}$ and $\mathbf{Q}_{t+1|t}$ follows a GARCH-type dynamic equation

$$\mathbf{Q}_{t+1|t} = \left(1 - \sum_{j=1}^J \alpha_j - \sum_{k=1}^K \beta_k \right) \bar{\mathbf{Q}} + \sum_{j=1}^J \alpha_j \varepsilon_{t+1-j} \varepsilon'_{t+1-j} + \sum_{k=1}^K \beta_k \mathbf{Q}_{t+1-k|t-k}.$$

As in the univariate case, α_j measures the news impact of past shocks, while β_k captures persistence in correlations.

Scale-location models capture dynamics in the return distribution through movements in the first two moments (mean and variance). A more general approach is provided by modeling changes in the dependencies between returns through their copulas. In particular, under weak conditions, it follows from Sklar's (1959) theorem that the joint return distribution of $\mathbf{r}_{t+1} = (r_{1t+1}, r_{2t+1}, \dots, r_{nt+1})$ can be decomposed into its univariate marginal distributions $P_1(r_{1t+1}), \dots, P_n(r_{nt+1})$ and a copula, C , that incorporates the joint distribution of the n marginal distributions:⁶

$$P(\mathbf{r}_{t+1}) = C [P_1(r_{1t+1}), \dots, P_n(r_{nt+1})]. \quad 66.$$

In a predictive context, it is appropriate to use a conditional copula representation of the form (Patton 2013)

$$P_{t+1|t}(\mathbf{r}_{t+1}) = C_{t+1|t} [P_{1t+1|t}(r_{1t+1}), \dots, P_{nt+1|t}(r_{nt+1})]. \quad 67.$$

To estimate copula models, one can adopt a two-stage approach. First, one estimates the conditional marginal distributions $P_{jt+1|t}(r_{jt+1})$, $j = 1, \dots, n$, using, e.g., a GARCH model; second, one uses the resulting probability integral transforms $P_{jt+1|t}(r_{jt+1})$ as inputs into the conditional copula model $C_{t+1|t}$.

5.4. Density Forecasting

Forecasts of the entire probability distribution of the outcome—density forecasts—are becoming increasingly common to use. The simplest class of models is the conditional scale-location models

⁶Copulas have some distinct advantages, such as being invariant under increasing and continuous transformations of the marginal distributions.

in Equation 62, which only require modeling the conditional mean and volatility functions, at least under the assumption of Gaussian shocks. Within this class of models, GARCH and Markov switching specifications such as Equations 48 and 50 have been fairly dominant, although stochastic volatility models such as Equation 60 are also popular.

The assumption of (conditionally) Gaussian innovations can, of course, easily be relaxed. For example, one can use a skewed t -distribution (Hansen 1994) to capture skewness and fat tails. Semiparametric density methods have also been proposed (see, e.g., Gallant & Tauchen 1989). These can use Hermite polynomials to add flexibility to a conditional scale-location model and take the form

$$p_{t+1|t}(r_{t+1}) = \frac{1}{c \sigma_{t+1|t}} \left(\sum_{i=0}^m \omega_i \eta_{t+1|t}^i \right)^2 \phi(\eta_{t+1|t}),$$

where $\phi(\cdot)$ is the standard Gaussian density function, $\eta_{t+1|t} = (r_{t+1} - \mu_{t+1|t})/\sigma_{t+1|t}$ is the standardized residual, and $c = \int \left(\sum_{i=0}^m \omega_i \eta_{t+1|t}^i \right)^2 \phi(\eta_{t+1|t}) d\eta_{t+1|t}$ is a normalization factor that ensures that the density integrates to one. When $m = 0$, the standard Gaussian density is recovered as a special case, while higher values of m add flexibility to the density specification.

5.4.1. Bayesian approaches. The Bayesian approaches that are typically used require fully specifying the underlying forecasting model and therefore generate a predictive density as part of the analysis. For example, this is the case for the stochastic volatility model in Equation 61.

Density forecasting is another area in which forecast combination has found good use. Geweke & Amisano (2011) propose constructing optimal pools of density forecasts. Let $p_{it|t-1}$ be the conditional density forecast of y_t from the i -th model, given information at time $t - 1$. Given a sample of data, $\{y_t\}$, $t = 1, \dots, T$, Geweke & Amisano propose choosing weights $\omega = (\omega_1, \dots, \omega_n)$ so as to maximize the sample log predictive score function

$$\omega = \arg \max_{\omega} T^{-1} \sum_{t=1}^T \ln \left(\sum_{i=1}^n \omega_i p_{it|t-1}(y_t) \right). \quad 68.$$

Geweke & Amisano consider combinations of a variety of GARCH and hidden Markov normal mixture models (corresponding to different specifications, $p_{it|t-1}$) and find that the optimal pool often yields interior solutions to Equation 68 that put considerable weight on at least two different density models, as opposed to assigning all or almost all weight to a single model.

Bayesian model averaging (BMA) is another useful technique that can be used to combine density forecasts. The BMA density forecast can be computed as a weighted average of n individual density forecasts

$$p^{\text{BMA}}(y_{t+1}|\mathbf{Z}_t) = \sum_{i=1}^n p(y_{t+1}|M_i, \mathbf{Z}_t) p(M_i|\mathbf{Z}_t), \quad 69.$$

where $p(M_i|\mathbf{Z}_t)$ is the posterior probability for model i given data $\mathbf{Z}_t$. Using Bayes' rule, this, in turn, can be computed from

$$p(M_i|\mathbf{Z}_t) = \frac{p(\mathbf{Z}_t|M_i)p(M_i)}{\sum_{j=1}^n p(\mathbf{Z}_t|M_j)p(M_j)}, \quad 70.$$

where $p(M_i)$ is the prior probability of (density) model M_i , while $p(\mathbf{Z}_t|M_i)$ is known as the marginal likelihood of model i . This is given by

$$p(\mathbf{Z}_t|M_i) = \int p(\mathbf{Z}_t|\theta_i, M_i) p(\theta_i|M_i) d\theta_i, \quad 71.$$

where $p(\theta_i|M_i)$ is the prior density of model i 's parameters and $p(\mathbf{Z}_i|\theta_i, M_i)$ is the likelihood of the data, given the parameters under the i -th model.

As can be seen from this description, the list of requirements for implementing BMA forecasts can be quite involved. First, one must have a set of models, $M_1, \dots, M_n$. For each of these models, one must be able to compute $p(M_i|\mathbf{Z}_i)$ in Equation 71, which can be time consuming for many types of models. Prior model probabilities, $p(M_1), \dots, p(M_n)$, are also needed—these are often set to $1/n$ —as are priors for the model parameters for each of the models, $p(\theta_1|M_1), \dots, p(\theta_n|M_n)$.⁷

5.4.2. Extracting densities from option prices. Market prices of derivatives such as options provide a special opportunity to obtain density estimates. Specifically, using a cross section of options with identical expiration date, T , but different strikes, X , one can compute option-implied density estimates. Let r_f denote the risk-free rate, while S_T is the price of the underlying asset at time T . Then the price of a call option at time $t < T$ is given by

$$C_t(T, X) = e^{-r_f(T-t)} \int_0^\infty (S_T - X, 0) f_t(S_T) dS_T, \quad 72.$$

where $f_t(S_T)$ is the option-implied risk-neutral density for S_T , and $F_t(S_T)$ is the corresponding cumulative distribution function. An estimate of $f_t(S_T)$ can be obtained by differentiating the expression for $C_t(T, X)$:

$$f_t(S_T) = e^{r_f(T-t)} \frac{\partial^2 C_t(T, X)}{\partial X^2} \Big|_{X=S_T}. \quad 73.$$

A variety of approximation methods exist for estimating $f_t(S_T)$, e.g., using a butterfly spread of options with neighboring strike prices:

$$f_t(X_n) \approx e^{r_f(T-t)} \left(\frac{C_t(T, X_{n+1}) - 2C_t(T, X_n) + C_t(T, X_{n-1})}{(\Delta X)^2} \right). \quad 74.$$

Complications arise when using data from options markets in this manner. First, the above formulas allow us to estimate the risk-neutral densities from option prices. Such densities are different from the density under the physical measure. Under assumptions about the process driving the underlying asset price and investor risk aversion (in incomplete markets), formulas can be derived that link the physical and risk-neutral probability distributions (for a thorough review and discussion of this topic, see Christoffersen, Jacobs & Chang 2013). Second, options on many securities are not very liquid, making it difficult to get a broad cross section from which option-implied densities can be estimated from, e.g., Equation 73. Measurement errors due to market microstructure effects and illiquid markets can also introduce biases in implied volatility estimates (Christensen & Prabhala 1998, Poon & Granger 2003). For a comparison of option-implied volatility to ARCH volatility models, see Donaldson & Kamstra (2005).

6. EVALUATING FINANCIAL FORECASTS

Evaluation of forecasts plays an important role in finance due to the presence of well-defined utility or loss functions that can be used to assess predictive accuracy. Another factor that plays a role is the relative ease with which financial forecasts can be exploited in investment strategies. Whereas it can be difficult to measure the impact on economic welfare of small improvements in the accuracy of

⁷Avramov (2002) applies BMA to a combination of 2¹⁴ possible model specifications obtained as the exhaustive set of combinations of 14 different predictors. Avramov (2002) computes the economic (utility) loss from ignoring model uncertainty and finds that this can be sizeable. He also finds that BMA forecasts are more robust than forecasts from individual models.

macroeconomic forecasts, it is usually easier to compute the effect of increased predictive accuracy on trading profits. The conventional tool for doing so is what is usually referred to as back tests, i.e., simulated real-time trading results based on some financial forecasting approach. There is also a literature on evaluation of volatility forecasts, which we do not focus on here (see, e.g., Patton 2011).

6.1. Utility-Based Loss Functions

Aït-Sahalia & Brandt (2001) discuss how determining an investor's optimal portfolio weights can be viewed as a prediction problem given the investor's utility function, u ; a budget constraint relating future wealth, $W_{t+1} = \omega'_t \mathbf{R}_{t+1}$, to portfolio weights, ω_t ; and a vector of gross asset returns, $\mathbf{R}_{t+1}$ (where, for simplicity, we assume an initial wealth of $W_t = 1$).

Suppose the investor chooses portfolio weights, ω_t , to maximize

$$\omega_t^* = \arg \max_{\omega_t} E_t[u(\omega'_t \mathbf{R}_{t+1})], \quad 75.$$

where $\omega'_t \mathbf{1} = 1$ for $\mathbf{1} = (1, 1, \dots, 1)'$. Assuming mean-variance preferences with a coefficient of absolute risk aversion $A \geq 0$,

$$E_t[u(W_{t+1})] = E_t[W_{t+1}] - \frac{A}{2} \text{Var}_t(W_{t+1}),$$

the optimal portfolio weights are a nonlinear function of the first and second moments of the wealth distribution, μ_t and Σ_t (Aït-Sahalia & Brandt 2001):⁸

$$\omega_t = \Sigma_t^{-1} \mathbf{1} \frac{AW_t - \mathbf{1}' \Sigma_t^{-1} \mu_t}{AW_t \mathbf{1}' \Sigma_t^{-1} \mathbf{1}} + \frac{\Sigma_t^{-1} \mu_t}{AW_t}. \quad 76.$$

To evaluate Equation 76, one option is to use plug-in estimates of μ_t and Σ_t . This is not a very desirable strategy, however, as it ignores estimation error. A more desirable strategy is to parameterize the investor's portfolio choice $\omega(\mathbf{x}_t, \beta)$ as a function of a set of predictors known at time t , $\mathbf{x}_t$, and parameters, β .

To keep the solution feasible and allow for multiple predictor variables, Aït-Sahalia & Brandt (2001) propose using a single-index specification $\omega(\mathbf{x}_t, \beta) = \omega(\mathbf{x}'_t \beta)$. Deriving the first-order condition from the optimization problem in Equation 75, we have

$$E \left[u' \left\{ W_t [\omega(\mathbf{x}'_t \beta) \mathbf{R}_{t+1}] \mathbf{R}_{t+1} \right\} \mid \mathbf{x}_t \right] = 0. \quad 77.$$

This equation can be used to estimate the parameters of the portfolio policy function by means of the GMM (generalized method of moments) estimation method, using functions of the data, $g(\mathbf{x}_t)$, to convert Equation 77 into sample moment restrictions.

In situations where ω_t is of sufficiently low dimension to allow for a grid search, simple numerical methods can be used to determine the investor's optimal portfolio weights given a predictive distribution of returns, $p(\mathbf{R}_{t+1} \mid \Omega_t)$. To see how this works, consider the optimal asset allocation of an investor with a power utility function who can choose between a risk-free asset and a single risky stock market portfolio with gross returns of $R_{f,t+1}$ and R_{t+1} , respectively:

$$U(\omega_t, R_{t+1}) = \frac{[(1 - \omega_t)R_{f,t+1} + \omega_t R_{t+1}]^{1-A}}{1 - A}, \quad 78.$$

⁸This expression assumes there is no risk-free asset available.

where A is the investor's coefficient of relative risk aversion. The investor chooses ω_t to solve the optimal asset allocation problem

$$\omega_t^* = \arg \max_{\omega_t} \int U(\omega_t, R_{t+1}) p(R_{t+1} | \Omega_t) dR_{t+1}. \quad 79.$$

The integral in Equation 79 can be numerically approximated using J draws from the predictive distribution

$$\hat{\omega}_{t,i} = \arg \max_{\omega_t} \frac{1}{J} \sum_{j=1}^J \left\{ \frac{[(1 - \omega_t)R_{f,t+1} + \omega_t R_{t+1}]^{1-A}}{1 - A} \right\}. \quad 80.$$

For example, in a frequentist setting, a GARCH model could be used as the predictive distribution from which to make draws. This procedure would typically condition on the parameter estimates. In a Bayesian setting, one could draw from the posterior predictive distribution, such as in Equation 61, to account for parameter uncertainty.

6.2. Economic Versus Statistical Loss

Using economic rather than purely statistical loss functions in the forecast evaluation step can make a substantial difference. Kandel & Stambaugh (1996) provide a range of illustrative examples showing situations where seemingly weak return predictability can have important economic consequences, e.g., in portfolio allocation problems.

Cenesizoglu & Timmermann (2012) empirically consider the performance of a range of forecasting models with time-varying mean and volatility. The parameters of all models are estimated recursively, and forecasting performance is measured out-of-sample. For the economic loss function, the forecasts are used to select optimal portfolio weights either under mean-variance preferences or under power utility, and performance is evaluated using these utility functions.

Cenesizoglu & Timmermann (2012) find that it is common for return prediction models to produce higher out-of-sample forecast MSEs than a model with a constant equity premium, yet simultaneously add economic value when their forecasts are used to guide portfolio decisions. Certainly, there is generally a positive correlation between a return prediction model's out-of-sample MSE performance and its ability to add economic value. However, this relation tends to be weak and explains a only small part of the cross-sectional variation in different models' ability to generate economic value in out-of-sample analysis.

These results suggest that underperformance along conventional measures of forecasting performance, such as root mean squared forecast errors, contain only limited information on whether return prediction models that allow for a time-varying mean or variance will help or hurt investors when these return forecasts are used to guide portfolio decisions.

7. CONCLUSION

As the cost of fitting an ever-larger number of increasingly sophisticated and flexible forecasting models to financial data sets declines, the ability of researchers to separate spurious patterns from genuine predictability becomes harder. Recent advances in dealing with the multiple hypothesis testing problem underlying the data mining problem offer some promise, although they often require revisiting the entire list of predictors that have been considered in the literature (as opposed to considering individual predictors in isolation), which can be a costly exercise for individual researchers. (For a study of technical trading rules, see Sullivan, Timmermann & White 1999; for

a study of factors explaining cross-sectional variation in expected returns, see Harvey, Liu & Zhu 2016.)

We are likely to see, in future research, a close contest between increasingly powerful forecasting methods with the ability to find predictive patterns even in settings with low signal-to-noise ratios and more sophisticated methods for dealing with overfitting and data mining. Ideally, economic theory will play an important role in this contest, helping researchers to identify robust predictability patterns, but also bearing in mind that sometimes predictive modeling is ahead of theory and that it may not always be clear how to pinpoint from theory which variables should possess predictive power over the outcome of interest.

Data limitations reduce the accuracy of financial forecasts in many situations. For example, our ability to predict variation in asset returns at business-cycle frequency using, e.g., slow-moving predictors such as interest-rate spreads or valuation ratios is ultimately limited by the small number of business cycles observed in historical data. Similarly, the tendency of predictable patterns in returns to self-destruct as a result of competitive pressures means that there are limits on how informative historical data will be for predicting future returns.

These points make many financial forecasting problems both challenging and fascinating. However, because the potential payoffs from uncovering even small degrees of predictability are so high, finance will undoubtedly remain an innovative and fast-paced arena for developing and testing new forecasting methods.

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LITERATURE CITED

- Aït-Sahalia Y, Brandt MW. 2001. Variable selection for portfolio choice. *J. Finance* 56:1297–351
- Aït-Sahalia Y, Mykland PA, Zhang L. 2005. How often to sample a continuous-time process in the presence of market microstructure noise. *Rev. Financ. Stud.* 18.2:351–416
- Andersen TG, Bollerslev T, Christoffersen PF, Diebold FX. 2006. Volatility and correlation forecasting. *Handb. Econ. Forecast.* 1:777–878
- Andersen TG, Bollerslev T, Diebold FX. 2007. Roughing it up: including jump components in the measurement, modeling, and forecasting of return volatility. *Rev. Econ. Stat.* 89(4):701–20
- Andreou E, Ghysels E, Kourtellis A. 2011. Forecasting with mixed-frequency data. In *The Oxford Handbook of Economic Forecasting*, ed. MP Clements, DF Hendry, pp. 225–45. Oxford, UK: Oxford Univ. Press
- Ang A, Piazzesi M. 2003. A no-arbitrage vector autoregression of term structure dynamics with macroeconomic and latent variables. *J. Monet. Econ.* 50(4):745–87
- Avramov D. 2002. Stock return predictability and model uncertainty. *J. Financ. Econ.* 64:423–58
- Blair BJ, Poon S-H, Taylor SJ. 2001. Forecasting S&P 100 volatility: the incremental information content of implied volatilities and high-frequency index returns. *J. Econom.* 105(1):5–26
- Bollerslev T. 1986. Generalized autoregressive conditional heteroskedasticity. *J. Econom.* 31:307–27
- Campbell JY, Shiller RJ. 1988. The dividend-price ratio and expectations of future dividends and discount factors. *Rev. Financ. Stud.* 1(3):195–228
- Campbell JY, Thompson SB. 2008. Predicting excess stock returns out of sample: Can anything beat the historical average? *Rev. Financ. Stud.* 21(4):1509–31

- Cenesizoglu T, Timmermann A. 2012. Do return prediction models add economic value? *J. Bank. Finance* 36(11):2974–87
- Chib S. 1998. Estimation and comparison of multiple change point models. *J. Econom.* 86:221–41
- Christensen BJ, Prabhala NR. 1998. The relation between implied and realized volatility. *J. Financ. Econ.* 50(2):125–50
- Christoffersen P, Jacobs K, Chang B. 2013. Forecasting with option-implied information. In *Handbook of Economic Forecasting*, Vol. 2A, ed. G Elliott, A Timmermann, pp. 581–656. Amsterdam: Elsevier
- Christopherson J, Ferson W, Glassman D. 1998. Conditioning manager alphas on economic information: another look at the persistence of performance. *Rev. Financ. Stud.* 11:111–42
- Clemen RT. 1989. Combining forecasts: a review and annotated bibliography. *Int. J. Forecast.* 5(4):559–83
- Cochrane JH. 2005. *Asset Pricing*. Princeton, NJ: Princeton Univ. Press. Revis. ed.
- Corsi F. 2009. A simple approximate long-memory model of realized volatility. *J. Financ. Econom.* 7(2):174–96
- Dangl T, Halling M. 2012. Predictive regressions with time-varying coefficients. *J. Financ. Econ.* 106:157–81
- DeMiguel V, Garlappi L, Uppal R. 2007. Optimal versus naive diversification: How inefficient is the 1/N portfolio strategy? *Rev. Financ. Stud.* 22(5):1915–53
- Diebold FX, Mariano RS. 1995. Comparing predictive accuracy. *J. Bus. Econ. Stat.* 13:253–63
- Donaldson RG, Kamstra M. 2005. Volatility forecasts, trading volume and the ARCH vs option-implied volatility trade-off. *J. Financ. Res.* 27:519–38
- Elliott G, Gargano A, Timmermann A. 2013. Complete subset regressions. *J. Econom.* 177:357–73
- Elliott G, Gargano A, Timmermann A. 2015. Complete subset regressions with large-dimensional sets of predictors. *J. Econ. Dyn. Control* 54:86–110
- Engle RF. 1982. Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica* 50:987–1007
- Engle RF. 2002. Dynamic conditional correlation: a simple class of multivariate generalized autoregressive conditional heteroskedasticity models. *J. Bus. Econ. Stat.* 20:339–50
- Engle RF, Sheppard K. 2001. *Theoretical and empirical properties of dynamic conditional correlation multivariate GARCH*. NBER Work. Pap. 8554
- Farmer L, Schmidt L, Timmermann A. 2017. *Pockets of predictability*. Unpublished manuscript
- Ferreira MA, Santa-Clara P. 2011. Forecasting stock market returns: The sum of the parts is more than the whole. *J. Financ. Econ.* 100(3):514–37
- Ferson WE, Sarkissian S, Simin TT. 2003. Spurious regressions in financial economics? *J. Finance* 58(4):1393–413
- Ferson WE, Schadt RW. 1996. Measuring fund strategy and performance in changing economic conditions. *J. Finance* 51:425–61
- Gallant AR, Tauchen G. 1989. Semiparametric estimation of conditionally constrained heterogeneous processes: asset pricing applications. *Econometrica* 57:1091–120
- Geweke J, Amisano G. 2011. Optimal prediction pools. *J. Econom.* 164:130–41
- Ghysels E, Sinko A, Valkanov R. 2007. MIDAS regressions: further results and new directions. *Econom. Rev.* 26:53–90
- Giacomini R, Rossi B. 2009. Detecting and predicting forecast breakdowns. *Rev. Econ. Stud.* 76:669–705
- Giacomini R, Rossi B. 2010. Forecast comparisons in unstable environments. *J. Appl. Econom.* 25:595–620
- Glosten LR, Jagannathan R, Runkle DE. 1993. On the relation between the expected value and the volatility of the nominal excess return on stocks. *J. Finance* 48(5):1779–801
- Hamilton JD. 1989. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica* 57:357–84
- Hansen BE. 1994. Autoregressive conditional density estimation. *Int. Econ. Rev.* 35:705–30
- Hansen PR, Huang Z, Shek H. 2012. Realized GARCH: a joint model for returns and realized measures of volatility. *J. Appl. Econom.* 27:877–906
- Hansen PR, Lunde A. 2005. A forecast comparison of volatility models: Does anything beat a GARCH(1, 1)? *J. Appl. Econom.* 20:873–89
- Hansen PR, Lunde A. 2011. Forecasting volatility using high frequency data. In *The Oxford Handbook of Economic Forecasting*, ed. MP Clements, DF Hendry, pp. 525–56. Oxford, UK: Oxford Univ. Press

- Hansen PR, Lunde A, Nason JM. 2011. The model confidence set. *Econometrica* 79:453–97
- Hansen PR, Timmermann A. 2012. *Choice of sample split in out-of-sample forecast evaluation*. Unpublished manuscript
- Hansen PR, Timmermann A. 2015a. Comment. *J. Bus. Econ. Stat.* 33(1):17–21
- Hansen PR, Timmermann A. 2015b. Equivalence between out-of-sample forecast comparisons and Wald statistics. *Econometrica* 83(6):2485–505
- Harvey CR, Liu Y, Zhu H. 2016. ...And the cross-section of expected returns. *Rev. Financ. Stud.* 29(1):5–68
- Hastie T, Tibshirani R, Friedman J. 2009. *The Elements of Statistical Learning*. New York: Springer. 2nd ed.
- Henkel SJ, Martin JS, Nadari F. 2011. Time-varying short-horizon predictability. *J. Financ. Econ.* 99(3):560–80
- Inoue A, Kilian L. 2005. In-sample or out-of-sample tests of predictability: Which one should we use? *Econom. Rev.* 23:371–402
- Johannes M, Korteweg A, Polson N. 2014. Sequential learning, predictability, and optimal portfolio returns. *J. Finance* 69(2):611–44
- Kamstra M, Kramer LA, Levi MD. 2015. Seasonal variation in Treasury returns. *Crit. Finance Rev.* 4:45–115
- Kandel S, Stambaugh RF. 1996. On the predictability of stock returns: an asset-allocation perspective. *J. Finance* 51:385–424
- Kelly B, Pruitt S. 2013. Market expectations in the cross-section of present values. *J. Finance* 68(5):1721–56
- Kendall MG. 1954. Note on bias in the estimation of autocorrelation. *Biometrika* 41:403–4
- Kim S, Shephard N, Chib S. 1998. Stochastic volatility: likelihood inference and comparison with arch models. *Rev. Econ. Stud.* 65(3):361–93
- Ludvigson SC, Ng S. 2007. The empirical risk–return relation: a factor analysis approach. *J. Financ. Econ.* 83(1):171–222
- Maheu JM, McCurdy TH. 2009. Forecasting the long-run equity return distribution. *J. Bus. Econ. Stat.* 27:95–112
- Marriott FHC, Pope JA. 1954. Bias in the estimation of autocorrelations. *Biometrika* 41:390–402
- McLean RD, Pontiff J. 2016. Does academic research destroy stock return predictability? *J. Finance* 71(1):5–32
- Neely CJ, Rapach DE, Tu J, Zhou G. 2014. Forecasting the equity risk premium: the role of technical indicators. *Manag. Sci.* 60(7):1772–91
- Pástor L', Stambaugh RF. 2001. The equity premium and structural breaks. *J. Finance* 56:1207–45
- Pástor L', Stambaugh RF. 2009. Predictive systems: living with imperfect predictors. *J. Finance* 64(4):1583–628
- Patton AJ. 2011. Volatility forecast comparison using imperfect volatility proxies. *J. Econom.* 160(1):246–56
- Patton AJ. 2013. Copula methods for forecasting multivariate time series. *Handb. Econ. Forecast.* 2:899–960
- Patton AJ, Sheppard K. 2015. Good volatility, bad volatility: signed jumps and the persistence of volatility. *Rev. Econ. Stat.* 97(3):683–97
- Paye BS. 2012. 'Déjà vol': predictive regressions for aggregate stock market volatility using macroeconomic variables. *J. Financ. Econ.* 106(3):527–46
- Paye BS, Timmermann A. 2006. Instability of return prediction models. *J. Empir. Finance* 13:274–315
- Pesaran MH, Timmermann A. 2007. Selection of estimation window in the presence of breaks. *J. Econom.* 137:134–61
- Pettenuzzo D, Timmermann A. 2011. Predictability of stock returns and asset allocation under structural breaks. *J. Econom.* 164:60–78
- Pettenuzzo D, Timmermann A, Valkanov R. 2014. Forecasting stock returns under economic constraints. *J. Financ. Econ.* 114:517–53
- Pettenuzzo D, Timmermann A, Valkanov R. 2016. A MIDAS approach to modeling first and second moment dynamics. *J. Econom.* 193(2):315–34
- Poon S-H, Granger CWJ. 2003. Forecasting volatility in financial markets: a review. *J. Econ. Lit.* 41:478–539
- Rapach DE, Strauss JK, Zhou G. 2010. Out-of-sample equity premium prediction: combination forecasts and links to the real economy. *Rev. Financ. Stud.* 23(2):821–62
- Rapach DE, Wohar ME. 2006. Structural breaks and predictive regression models of aggregate US stock returns. *J. Financ. Econom.* 4(2):238–74
- Romano J, Wolf M. 2005. Stepwise multiple testing as formalized data snooping. *Econometrica* 73(4):1237–82
- Rossi A, Timmermann A. 2015. Modeling covariance risk in Merton's ICAPM. *Rev. Financ. Stud.* 28:1428–61

- Rossi B. 2013a. Advances in forecasting under instability. In *Handbook of Economic Forecasting*, Vol. 2B, ed. G Elliott, A Timmermann, pp. 1203–324. Amsterdam: Elsevier
- Rossi B. 2013b. Exchange rate predictability. *J. Econ. Lit.* 51:1063–119
- Sarno L, Schneider P, Wagner C. 2016. The economic value of predicting bond risk premia. *J. Empir. Finance* 37:247–67
- Sklar A. 1959. Fonctions de répartition à n dimensions et leurs marges. *Publ. Inst. Stat. Univ. Paris* 8:229–31
- Smith S, Timmermann A. 2017. *Detecting breaks in real time: a panel forecasting approach*. Unpublished manuscript
- Stambaugh RF. 1999. Predictive regressions. *J. Financ. Econ.* 54:375–421
- Sullivan R, Timmermann A, White H. 1999. Data-snooping, technical trading rule performance, and the bootstrap. *J. Finance* 54(5):1647–91
- Timmermann A. 2006. Forecast combinations. *Handb. Econ. Forecast.* 1:135–96
- van Binsbergen JH, Koijen R. 2010. Predictive regressions: a present-value approach. *J. Finance* 65(4):1439–71
- Welch I, Goyal A. 2008. A comprehensive look at the empirical performance of equity premium prediction. *Rev. Financ. Stud.* 21(4):1455–508
- White H. 2000. A reality check for data snooping. *Econometrica* 68:1097–126
- Zhou G. 2010. How much stock return predictability can we expect from an asset pricing model? *Econ. Lett.* 108:184–86